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ELIMINATE YELLOW FEVER EPIDEMICS

Root cause analysis of a yellow fever outbreak

User manual and questionnaires

World Health Organization

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Abbreviations and acronyms

CDC	United States Centers for Disease Control and Prevention
EPI	Essential Programme on Immunization
TDR	UNICEF/UNDP/World Bank/WHO Special Programme for Research and Training on Tropical Diseases
UNDP	United Nations Development Programme
UNICEF	United Nations Children's Fund
USAID	United States Agency for International Development
WHO	World Health Organization

Glossary of terms

Community: a group of people connected by common characteristics, such as geographic location, age, gender, profession, ethnicity, faith, shared vulnerability or risk, or shared interests and values *(1)*.

Preventive mass vaccination campaign: a strategy to rapidly raise population immunity and reduce the number of susceptible individuals to achieve disease control or elimination goals.

Stakeholders: governments and community leaders that have a vested interest in protecting the health of their own country, region or community *(1)*.

Vaccine hesitancy: delay in acceptance or refusal of safe vaccine despite availability of vaccination *(2)*.

Reactive vaccination campaign: a strategy implemented in response to an outbreak of a vaccine-preventable disease to rapidly raise immunity in a specific population or area where the disease has been detected and to control its spread.

Yellow fever immunity gap: a reduction in population-wide immunity to yellow fever. This can occur when a significant portion of the population has not been exposed to the yellow fever virus, either through natural infection or vaccination, leading to low herd immunity. The immunity gap can be estimated with the aid of the immunity gap analysis tool (WHO, unpublished).

Introduction

Since 2020, there has been a resurgence of yellow fever outbreaks in endemic countries of Africa. This resurgence predominantly affects countries with a history of mass vaccination campaigns where routine immunization coverage could not sustain the gains from these campaigns. Additionally, vulnerable, migrant, and hard-to-reach populations were often missed. A yellow fever outbreak provides an opportunity to learn more about the local epidemiology of the disease by identifying its source and contributing factors. It also helps strengthen the immunization program by uncovering the causes of underlying immunity gaps.

The Root Cause Analysis of a Yellow Fever Outbreak tool aims to facilitate an in-depth analysis of yellow fever epidemics in endemic countries, including the underlying causes of immunity gaps. This analysis is expected to contribute to capacity building in yellow fever epidemiology and vaccination. Besides offering an initial understanding of how and why the outbreak occurred, the assessment results can be used to develop a district-tailored recovery plan for vaccination services. By investigating the reasons for failure to vaccinate and/or vaccine failure, including provider and user-based reasons, the findings will ultimately contribute to strengthening the routine immunization program.

1. Purpose of the tool

The tool for the root cause analysis of yellow fever outbreaks aims at facilitating an in-depth analysis of underlying factors that contribute to yellow fever epidemics. It is intended to provide a foundation for improving our understanding of yellow fever epidemiology and yellow fever immunity gaps in high-risk countries, as well as contributing to capacity building on yellow fever epidemiology and vaccination.

2. Methodology development

Literature review

We conducted a scoping review of the quantitative and qualitative studies reporting on yellow fever burden and epidemiology, vaccination, or factors leading to yellow fever transmission and outbreak. Our database search included PubMed, that combined subject heading searches with free-text keyword searches. We did not restrict our review to a specific publication period or geographic location of the outbreaks. The included studies were published in English and French, or in other languages with English abstracts. Additionally, we reviewed the reference lists of relevant studies to identify further research. This review included both published and unpublished articles. We also reviewed World Health Organization (WHO) guidelines on yellow fever, internal tools, weekly epidemiological reports, WHO arboviruses and immunization guidelines, national investigation reports, and instructions from each WHO pre-qualified yellow fever vaccine manufacturer. For the root cause analysis method, we reviewed internal documents and root cause analysis guidelines. The list of published documents reviewed is provided in Annex 1.

Pilot testing

A first version of the tool was piloted in three yellow fever high-risk countries fulfilling two criteria: history of mass vaccination campaigns and resurgence of yellow fever cases requiring a reactive vaccination campaign in 2022.[1] The three countries were: Guinea (September-October 2023); Cameroon (November-December 2023); and the Central African Republic (March-April 2024).

Piloting the tool in these settings provided an opportunity to assess its completeness, clarity, and usability. Field testing helped identify missing questions or response options, verify that the sequence of questions was logical and easy for respondents to follow, and clarify wording that could be interpreted ambiguously. It also enabled the reformulation or removal of items that did not yield meaningful information for the analysis.

Peer review

A second version of the tool was peer-reviewed after the pilot testing phase by experts from the Eliminate Yellow fever Epidemics vaccine delivery working group and risk analysis working group, the WHO department of Immunization, Vaccines and Biologicals, the United States Centres for Disease Control and Prevention (CDC) and the Africa Centers for Disease Control and Prevention.

Throughout the revision process, particular attention was given to ensuring consistent and clear terminology, and confirming that each question contributed effectively to the root-cause analysis. These considerations informed the final version and content of the toolkit.

All external experts who reviewed the toolkit submitted a WHO declaration of interest form, disclosing potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence concerning the subject.

WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meeting or covered by the guidance.

3. Target audience

The tool will be useful to all national actors and stakeholders involved in outbreak investigation and vaccine-preventable disease surveillance to improve understanding of the yellow fever epidemiological situation and to identify priorities for strengthening capacity relevant for outbreak prevention and investigation. It will also help the national Essential Programme on Immunization (EPI) to identify priorities for strengthening capacity in districts with yellow fever vaccination gaps and allow WHO country and regional offices to guide technical support to countries and identify priorities for strengthening capacity.

4. When to use the tool?

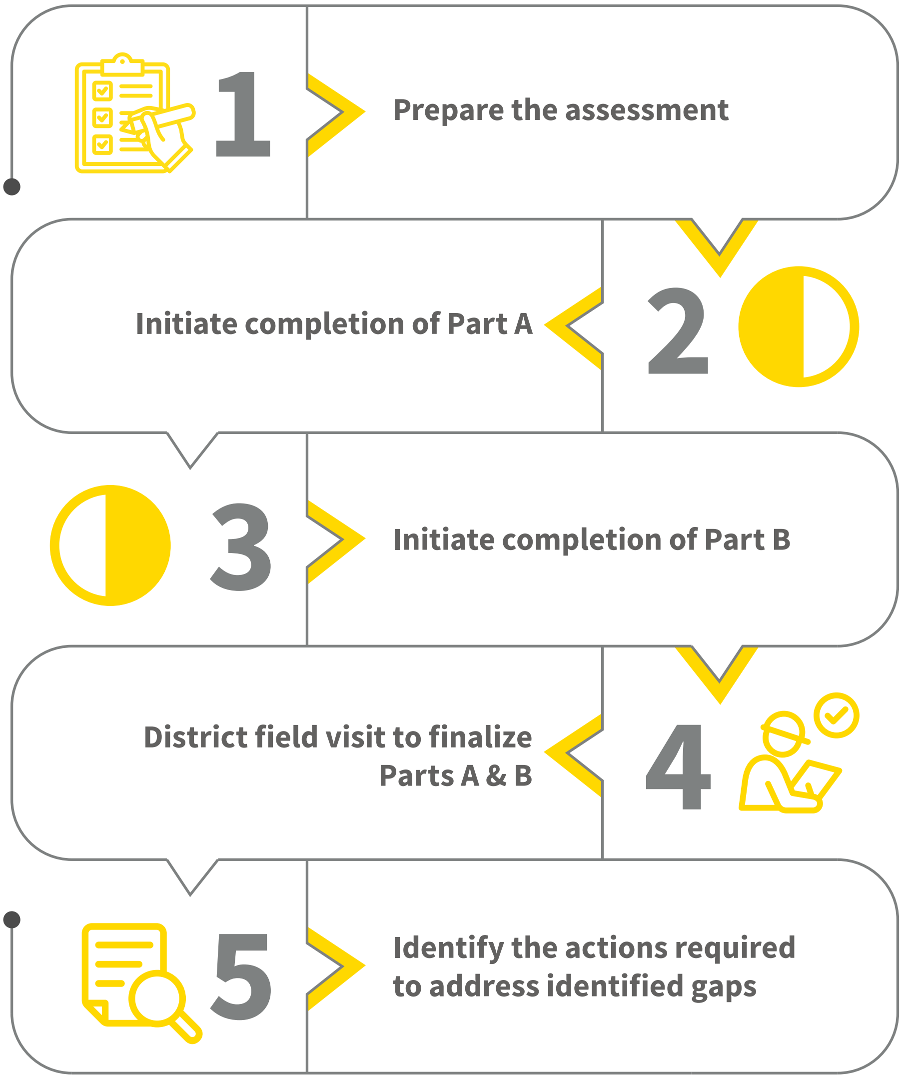
The assessment can be conducted in a district, which recently experienced a yellow fever outbreak that triggered a reactive vaccination campaign and where there is a need for an in-depth evaluation of root causes.

1 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(3)*.

5. How to use the tool?

The tool can be used through the two related documents, Part A and Part B (Fig. 1).

Fig. 1. Five-step procedure for the root cause analysis of yellow fever outbreaks.



Step 1. Preparing the assessment

Form a team

Officials from the ministry of health working closely with the WHO country office decide on the composition of the evaluation team. The team must be multidisciplinary with skills in outbreak investigation and vaccination activities (mass campaign and routine vaccination). It must be led by an international, experienced epidemiologist and include representatives of both the EPI and emergency programs and health personnel from the district and central levels. This is essential as information is required from all levels and also because they are responsible for implementing corrective actions. The use of this tool does not require specific additional training.

Estimate the required budget for the field assessment

In the preparatory stage, it is essential to accurately forecast the budget for the field assessment. Costs that need consideration include expenses for transport and fuel, expenses for data collection materials (such as pencils and questionnaire photocopies), and the fees for those conducting the assessment. Additional costs may be considered, subject to the context. A template to help budget planning for the investigation is provided in Annex 2.

Protect team members

All team members must provide proof that they have been vaccinated against yellow fever at least 10 days before the start of the field investigation.

Briefing

The ministry of health should hold a briefing session with the assessment team at the start, including those responsible for implementing the recommendations.

Step 2. Initiate completion of Part A (yellow fever outbreak in-depth analysis)

Part A of the tool is a checklist that provides a list of key data to collect to investigate underlying factors for a yellow fever outbreak. Apart from the description of the distribution of the epidemic in terms of time, place and individual case characteristics, the understanding of a yellow fever outbreak requires an understanding of the mode of transmission and a description of factors affecting yellow fever amplification. Data are collected retrospectively. Sources of information are provided for each topic assessed. Many source documents are likely to be available at central level. Source documents only available at district level can be consulted at the time of the field visit (Step 4).

Step 3. Initiate completion of Part B (tool for root causes of immunity gaps)

Part B of the tool is a questionnaire that facilitates the understanding of the reasons for the immunity gap in the district and is divided into the following two key sections.

- Section A (yellow fever vaccination coverage estimates) is designed to capture yellow fever vaccination coverage of routine activities and mass vaccination campaigns implemented in the district under investigation.
- Section B (reasons for immunity gaps) is designed to investigate the chain of causality of the immunity gap leading to the yellow fever outbreak. The chain of causality is determined using a phased approach, consistent with the approach detailed in the WHO measles outbreak root cause analysis guide 2021 *(4)*. The underlying causes of an immunity gap that may have led to a yellow fever outbreak are listed in Table 1.

Table 1. Immediate and underlying causes related to a yellow fever outbreak in a district

<table>
<tr><th colspan="4">Failure to vaccinate</th></tr>
<tr><th colspan="2">Reasons</th><th>Immediate causes</th><th>Underlying causes</th></tr>
<tr><td colspan="2">Policy-based</td><td>• Inaccuracy of estimation of yellow fever vaccination coverage
• Gap in the vaccination plan for the general population and population at higher risk for yellow fever</td><td></td></tr>
<tr><td rowspan="2">Provider-based</td><td>Routine activities</td><td>• Missed routine vaccination sessions
• Stock-out of yellow fever vaccines/vaccination equipment
• Lack of qualified vaccinators
• Lack of knowledge/misconceptions regarding the yellow fever vaccination policy
• Lack of knowledge/misconceptions regarding the co-administration of the yellow fever vaccine with another antigen
• Lack of knowledge/misconceptions regarding the contraindications of the yellow fever vaccine
• Other reasons for not being vaccinating against yellow fever</td><td>• Limitation in skills enhancement
• Lack of supervision in the vaccination activities</td></tr>
<tr><td>Mass vaccination campaigns</td><td>• Lack of knowledge/misconceptions regarding the co-administration of the yellow fever vaccine with another antigen
• Lack of knowledge/misconceptions regarding the contraindications of the yellow fever vaccine
• Other reasons for not being vaccinating against yellow fever:
▪ stock-out of vaccines
▪ district not fully covered by the vaccination campaign</td><td>• Limitation in skills enhancement
• Lack of supervision</td></tr>
<tr><td colspan="2">User-based</td><td>• Vaccines hesitancy/obstacles to vaccination activities (routine and campaigns):
▪ Yellow fever vaccines knowledge and misconceptions
▪ Obstacles to routine vaccination activities
▪ Lack of awareness of the mass vaccination campaigns</td><td></td></tr>
<tr><th colspan="4">Vaccine failure</th></tr>
<tr><th colspan="2">Reasons</th><th>Immediate causes</th><th>Underlying causes</th></tr>
<tr><td colspan="2">Provider-based</td><td>• Poor quality of yellow fever vaccines due to:
▪ defective ice-lined refrigerator
▪ poor storage of the yellow fever vaccine and/or diluents during and outside the vaccination sessions
• Staff practice failure in the injection</td><td>• Lack of trained staff
• Misconceptions/lack of knowledge</td></tr>
<tr><td colspan="2">User-based</td><td>• Primary failure
• Secondary failure
• Host factors</td><td></td></tr>
</table>

Sections are clickable for easier navigation

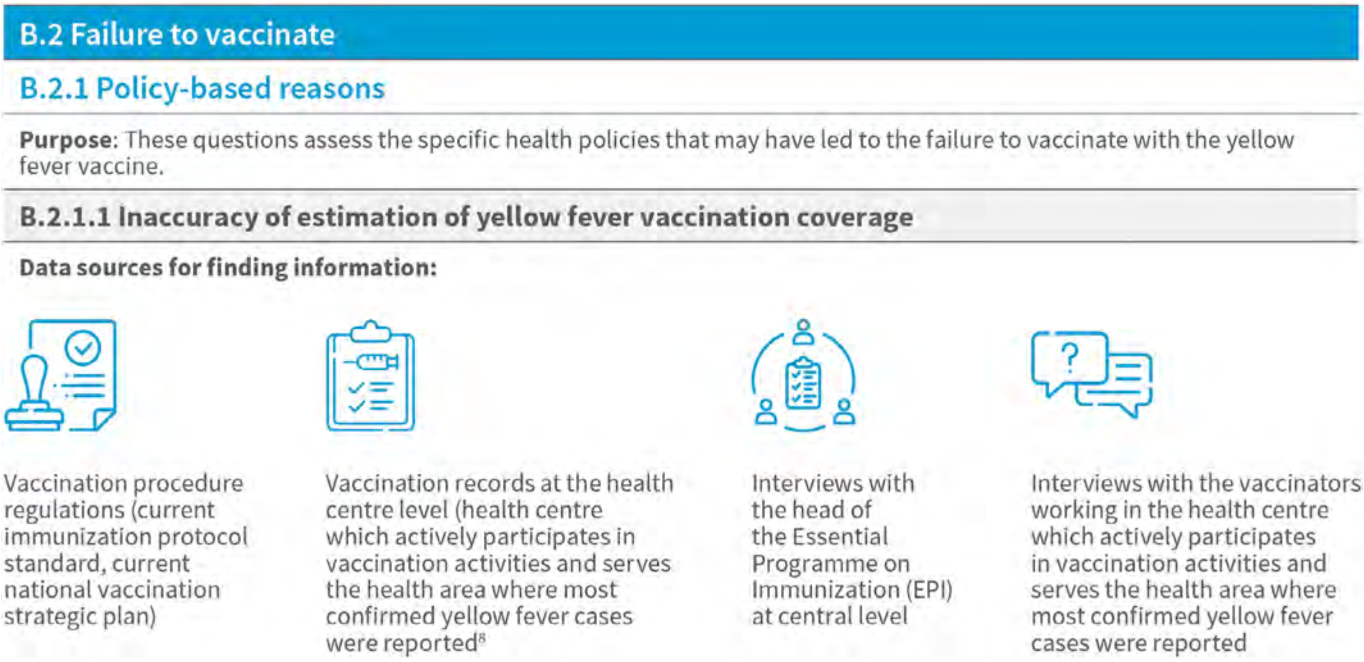
To initiate the analysis of the root causes of a yellow fever outbreak, data are first collected through a document review and one-on-one conversation at central level.

The document review should include:

1. documents shared by WHO;
2. documents shared by the country;
3. published and unpublished literature of yellow fever epidemiological information in the country.

A list of sources to find information that facilitates the recording of data is provided at the beginning of each section of the part B questionnaire under the section “data source” as shown below.

Box 1. Example of data sources for finding information



B.2 Failure to vaccinate

B.2.1 Policy-based reasons

Purpose: These questions assess the specific health policies that may have led to the failure to vaccinate with the yellow fever vaccine.

B.2.1.1 Inaccuracy of estimation of yellow fever vaccination coverage

Data sources for finding information:

Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)

Vaccination records at the health centre level (health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported[8]

Interviews with the head of the Essential Programme on Immunization (EPI) at central level

Interviews with the vaccinators working in the health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported

If the document review yields incomplete information, the data collection team is advised to collect information from the EPI focal point at central level for additional insights.

One-on-one conversation should be conducted with the EPI focal point and logistician of the vaccine storage room at the central level.

In the event that vaccine failure is identified as a cause of the yellow fever outbreak, observations of the vaccine store at central level are initiated.

The procedures used for data collection are indicated within the questionnaire before each question.

Important: In Step 3, the investigators proceed to the questions with previous mentions of DR (document review), DR and EPI C (document review and interview with the national EPI focal point), Log C (interview with the logistician of the vaccine storage room at the central level) and, in the event of vaccine failure, OBS C (observation of the vaccine store at central level).

Step 4. District and health centre field visit

Once information is gathered nationally, the assessment team completes Part A and Part B with further data collection in the district under investigation, as well as in the health centre actively participating in vaccination activities within that district and serving the health area where most confirmed yellow fever cases[2] *(3)* were reported.

Information collected in part B during the field visit derives from a variety of activities, including reviewing documents, conducting one-on-one conversations with health professionals or yellow fever case(s), or making structured observations.

The document review should include documents available at district and health facility levels. Each subsection of the Part B questionnaire starts with a mention of the documents required for the investigation.

The position of the personnel involved in the vaccination activities and interviewed by the data collection team is mentioned within the questionnaire prior to each question, as presented in Step 3.

In the event that vaccine failure is identified as a cause of the yellow fever outbreak, observations of the vaccine storage room at district and health centre levels are conducted after the observation completed in Step 3 at central level. Additionally, structured observations of vaccine activities are conducted, if circumstances allow.

Important: In Step 4, the team proceed to the questions with previous mentions of EPI D (interview with the EPI focal point at district level), Log D (interview with the logistician of the vaccine storage room at district level), EPI HC (interview with the EPI focal point at the health centre level), DIR (interview with the director of the health centre), CHW (Log HC (interview with the logistician of the vaccine storage room at the health centre level), and confirmed cases or their families.

In the event of vaccine failure, the team proceed to the questions with previous mentions of OBS D (observation of the vaccine store at district level) and OBS HC (observation of the vaccine store at health centre level).

Step 5. Identify actions to address identified gaps

The international, experienced epidemiologist leads the reporting of the results of the assessment. It is recommended to conduct a feedback workshop after the country visit to share initial findings. The session also provides an opportunity to incorporate and refine findings if needed. Participants in this workshop should include the evaluation team, national health authorities (entities involved in surveillance, outbreak investigation and immunization activities), the individuals who were interviewed during the on-site visit, country office representatives from WHO and UNICEF, and representatives from the WHO Regional Office for Africa.

The workshop should include a dedicated session on proposed measures to address root causes and barriers identified during the evaluation, with the aim to pinpoint necessary corrective measures and establish priorities. Following the online workshop discussions, the report is amended, and the final version is shared with the workshop attendees.

2 For the current yellow fever confirmed case definition, please refer to For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(3)*.

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Annex 1.

List of documents included in the literature review

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Annex 2.
Budget planning for the investigation

Tables A.2.1 to A.2.4 are intended to help plan a budget for the evaluation of the root causes of a yellow fever outbreak.

Table A.2.1. Staffing budget

Staff	Number	Number of days	Total	Rate (country currency)	
				Per day	Total
Epidemiologist					
EPI focal point at central level					
Head of the emergency program(s)					
WHO country office National Professional Officer					
Other personnel					
Subtotal					

Table A.2.2. Supplies budget

Items	Price/units (country currency)	Number of units	Total
Pencils			
Photocopies			
Other supplies			
Subtotal			

Table A.2.3. Transport budget

Items	Price/units (country currency)	Number of units	Total
Vehicle			
Fuel			
Subtotal			

Table A.2.4. Communication

Items	Price/units (country currency)	Number of units	Total
Telephone			
Other communication items			
Subtotal			

Evaluation total budget (country currency): ____

Annex 3.
Questionnaires for root cause analysis of a yellow fever outbreak

Questionnaires

These questionnaires are also available in a Word format as an annex to this user manual on the web.

Part A. Yellow fever outbreak in-depth analysis: assessment checklist

Objective

- To facilitate in-depth analysis of underlying factors of yellow fever epidemics.

Key considerations

- Part of the complexity of yellow fever epidemics is the multifactorial and evolving nature of risk and its inherent unknowns.
- Assessment of underlying reasons for a yellow fever outbreak requires a description of the outbreak in terms of time, place and person, an understanding of its mode of transmission, the likelihood of underestimation of yellow fever cases, and factors affecting disease transmission and propagation.
- The template form below should be used to consolidate and analyze available information about a yellow fever outbreak as soon as it is declared as ended by national health authorities. The form aims to collect additional information about the outbreak beyond routine reporting.
- The Part A assessment should be completed after the implementation of outbreak response measures (for example, after a reactive vaccination campaign). It should be completed by an epidemiologist and include health personnel from the district and central levels.
- Reasons for the unavailability of source documents will alert to areas in need of corrective measures to improve outbreak investigation capacity.
- Following completion of the **Part A** form, gaps in yellow fever immunity can be assessed using **Part B**.

Structure of the checklist

This checklist is divided into three components, which are then subdivided into a total of nine assessment areas covering all aspects of a yellow fever outbreak.

- Component 1. Outbreak description:
    - time
    - place
    - person
    - mode of transmission (for example, rural, urban, sylvatic).

- Component 2. Likelihood of underestimation of the scale of the outbreak:
  - surveillance indicators.
- Component 3. Factors affecting disease transmission and propagation:
  - vector
  - non-human primates
  - human exposure to sylvatic environment
  - level of protection of district population.

How to use the assessment checklist

- For each of the key data/indicator(s), use the suggested source document to obtain the information.
- Many source documents are available at national level; others will require a field visit to the affected district.
- Any "absent" source document should be recorded, and justifications provided.
- A report should be compiled to present all the findings.
- Technical reference documents:
  - Yellow fever outbreak toolbox. Geneva: World Health Organization; 2024 (Yellow fever Outbreak Toolbox (who.int), accessed 24 August 2024);
  - Yellow fever: vaccine-preventable diseases surveillance standards. Geneva: World Health Organization; 2020
  - (https://www.who.int/publications/m/item/vaccine-preventable-diseases-surveillance-standards-yellow-fever, accessed 24 August 2024);
  - Investigation of yellow fever epidemics in Africa: field guide. Geneva: World Health Organization; 2008 (https://iris.who.int/handle/10665/69874, accessed 24 August 2024).

Yellow fever outbreak in-depth analysis: assessment checklist	
Description of the affected district	
District information	
Evaluation date	
Name of the facility where the index case was notified	
City	
District	
Country	
Obtain/draw map of district showing potential endemic areas (forests with non-human primates) and areas with previous reports of cases	
District: population estimates (year, source), including mobile populations	
Record any population movements or gatherings in months preceding the yellow fever outbreak response	

Outbreak description

Time	Suggested source document
Obtain the date of onset of disease of the index case and date of onset of the last case.	*National line list of yellow fever cases*
Record the total number of probable and confirmed cases of yellow fever over the epidemic period.	*National line list of yellow fever cases*
Draw the epidemic curve.	*National line list of yellow fever cases*
Comments on source documents if unavailable (name and reason for unavailability):	

Place	Suggested source document
Map place of residency of probable and confirmed yellow fever cases.	*International Coordinating Group on Vaccine Provision request, national line list of yellow fever cases*
Describe zone affected (village, proximity to forest area, transport hubs and main roads).	
Comments on source documents if unavailable (name and reason for unavailability):	

Person	Suggested source document
Describe the individual case characteristics by calculating: • sex ratio; • percentage of cases per age group (<5 years, 5–45 years, 46–60 years); • age range (minimum, maximum).	*National line list of yellow fever cases*
Review profession of cases and report any occupational risk factors (for example, extraction industries such as mining or forestry, duties with high contact with sylvatic habitats such as agriculture or road construction).	*Case investigation reports*
Describe type of community affected by yellow fever (for example, internally displaced persons, seasonal workers, nomads, marginalized populations, etc.).	*Case investigation reports*
Comments on source documents if unavailable (name and reason for unavailability):	

Severity of outbreak	Suggested source document
Calculate: • overall case fatality rate • case fatality rate by sex • case fatality rate by age groups.	*National line list of yellow fever cases*
Check if outbreak was large/disruptive.[1]	*National line list of yellow fever cases and map of cases*
Comments on source documents if unavailable (name and reason for unavailability):	

1 Defined as five or more cases close in space and time requiring a response in endemic areas. Or >1 case in an area previously without yellow fever.

Mode of transmission	Suggested source document
Indicate type of setting of the likely place of infection of the index case and subsequent cases based on age, profession, forest activities, travel history during the two weeks preceding onset of symptoms, urban or rural setting.	*Case investigation reports*
Review if yellow fever cases (index case and subsequent) travelled to endemic zones (forest) or a zone where an outbreak was underway (specify, place, date and duration).	*Case investigation reports*
Check if yellow fever cases (index case and subsequent cases) were living in an urban, village or forest area.	*Case investigation reports*
Record vector species identified by field entomological investigation.	*Field entomological investigation report*
Indicate likely transmission pattern(s) involved during the outbreak (sylvatic, intermediate, urban).	
Comments on source documents if unavailable (name and reason for unavailability):	

Likelihood of underestimation of the outbreak

Surveillance indicators	Suggested source document
Check if the universal health coverage index score is ≥80%.[2]	link to data source
Assess if the population is scattered/check for district-level access to care indicator.	*Google Earth/Google map*
Check if community health workers were trained in detecting a suspected yellow fever case and involved in surveillance of yellow fever.	*Interview district health authorities*
Check care pathway of 2–3 most recent yellow fever cases and review reasons for eventual delays in seeking care at health facility.	*Patient file, interview clinician, interview*
Check in health facility register if other suspected yellow fever cases were identified, but not tested; if so, inquire about reason for not sampling the suspected case (s).	*Health facility register (where index case was notified)*
Check for completeness of reporting in the district or health area: percentage of designated units reporting suspected yellow fever cases (target ≥80%).	*Surveillance data at district health level*
Check specimen rate: percentage of suspected yellow fever cases with a specimen collected (target ≥80%)	*Health facility register (where index case was notified)*
Comments on source documents if unavailable (name and reason for unavailability):	

2 Coverage of essential health services (Sustainable Development Goal 3.8.1) is defined as the average coverage of essential services based on tracer interventions that include reproductive, maternal, newborn and child health, infectious diseases, non-communicable diseases and capacity and access to services among the general population and the most disadvantaged populations. If score is less than 80%, there is likely underdetection of yellow fever cases due to lack of access to essential health services. The indicator is measured as an index reported on a unitless scale from 0 to 100, which is calculated as the geometric mean of 14 tracer indicators of health service coverage. These indicators are intended to be indicative of service coverage and should not be interpreted as a complete or exhaustive list of health services or interventions needed to achieve universal health coverage.

Factors affecting disease transmission and propagation

Vector	Suggested source document
Check if routine entomological surveillance is functional in the district; if so, check the availability of routine surveillance data and the trend in yellow fever vector density and distribution.	*Interview Vector Control Division*
Record vector species identified around household cases and entomological indices.	*Field entomological investigation report*
Check for surveys on yellow fever vector distribution.	*Literature search (for example, PubMed), interview national entomologists*
Check if vector control measures were implemented before the yellow fever outbreak.	*Interview Vector Control Division and district health authorities*
Check for surveys on the insecticide resistance of yellow fever vectors.	*Interview Vector Control Division, literature search (for example, PubMed)*
Comments on source documents if unavailable (name and reason for unavailability):	

Vector	Suggested source document
Check if routine entomological surveillance is functional in the district; if so, check the availability of routine surveillance data and the trend in yellow fever vector density and distribution.	*Interview Vector Control Division*
Record vector species identified around household cases and entomological indices.	*Field entomological investigation report*
Check for surveys on yellow fever vector distribution.	*Literature search (for example, PubMed), interview national entomologists*
Check if vector control measures were implemented before the yellow fever outbreak.	*Interview Vector Control Division and district health authorities*
Check for surveys on the insecticide resistance of yellow fever vectors.	*Interview Vector Control Division, literature search (for example, PubMed)*
Comments on source documents if unavailable (name and reason for unavailability):	

Non-human primates	Suggested source document
Inquire about the presence of non-human primates in the affected district.	*Interview the One Health focal point and district representative of the ministry in charge of forestry*
Inquire about any change in the number and distribution of the non-human primate population in the district (if present).	*Interview the One Health focal point and district representative of the ministry in charge of forestry*
Comments on source documents if unavailable (name and reason for unavailability):	

Human exposure to sylvatic environment	Suggested source document
Inquire about migration or displacement of unprotected populations (workers, nomads, internally displaced persons, refugees, transient populations, etc.) in the district.	*Interview district authorities*
Inquire about deforestation in the district forest.	*Interview the district representative of the ministry in charge of forestry*
Inquire about changes in the local population's activities that could result in increased exposure to the forest environment.	*Interview district authorities*

Comments on source documents if unavailable (name and reason for unavailability):

Part B. Data collection tool for analysing root causes of immunity gaps

Abbreviations and acronyms

EPI	Essential Programme on Immunization
ILR	ice-lined refrigerator
MM/YYYY	month/year
PMVC	preventive mass vaccination campaign
RVC	reactive vaccination campaign
TDR	UNICEF/UNDP/World Bank/WHO Special Programme for Research and Training on Tropical Diseases
UNDP	United Nations Development Programme
UNICEF	United Nations International Children's Fund
USAID	United States Agency for International Development
WHO	World Health Organization

Glossary of terms

Community: a group of people connected by common characteristics, such as geographic location, age, gender, profession, ethnicity, faith, shared vulnerability or risk, or shared interests and values *(1)*.

Preventive mass vaccination campaign: a strategy to rapidly raise population immunity and reduce the number of susceptible individuals to achieve disease control or elimination goals.

Stakeholders: governments and community leaders that have a vested interest in protecting the health of their own country, region or community *(1)*.

Vaccine hesitancy: delay in acceptance or refusal of safe vaccine despite availability of vaccination *(2)*.

Reactive vaccination campaign: a strategy implemented in response to an outbreak of a vaccine-preventable disease to rapidly raise immunity in a specific population or area where the disease has been detected and to control its spread.

Yellow fever immunity gap: a reduction in population-wide immunity to yellow fever. This can occur when a significant portion of the population has not been exposed to the yellow fever virus, either through natural infection or vaccination, leading to low herd immunity. The immunity gap can be estimated with the aid of the immunity gap analysis tool (WHO, unpublished).

Questionnaire outline

Part B is divided into the following two key sections.

Section A (yellow fever vaccination coverage estimates in the district under investigation) is designed to report the estimated yellow fever vaccination coverage of the routine activities at central, district and health centre levels. It is also designed to determine whether yellow fever mass vaccination campaigns have been implemented in the district under investigation and, if confirmed, to report the estimated vaccination coverage.

Section B (reasons for immunity gaps) is designed to investigate the chain of causality of the immunity gap leading to the yellow fever outbreak. The chain of causality is determined using a phased approach, consistent with the approach detailed in the World Health Organization (WHO) measles outbreak root cause analysis guide published in 2022 *(3)*. The reasons for an immunity gap (referred to as "first-level causes") among confirmed cases[3] may be due to a failure to vaccinate when confirmed yellow fever cases are not vaccinated and/or a vaccine failure when confirmed yellow fever cases are vaccinated[4].

Further investigations delve into the second-level causes, followed by lower-level causes through a series of 'why' questions, where the answer to each 'why' question leads to another 'why' question that reveals the chain of causality down to the most fundamental causes. Thus, these investigations determine why people were not vaccinated and/or why the vaccine did not protect them, and whether these reasons were related to the national vaccination policy, the service provider or the user. These sections are then subdivided into subsections that delve deeper into the underlying causes.

The underlying causes of an immunity gap that may have led to a yellow fever outbreak are listed in Table 1.

3 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.
4 If a confirmed case reports to be vaccinated with no proof of vaccination, both failure to vaccinate and vaccine failure will be investigated.

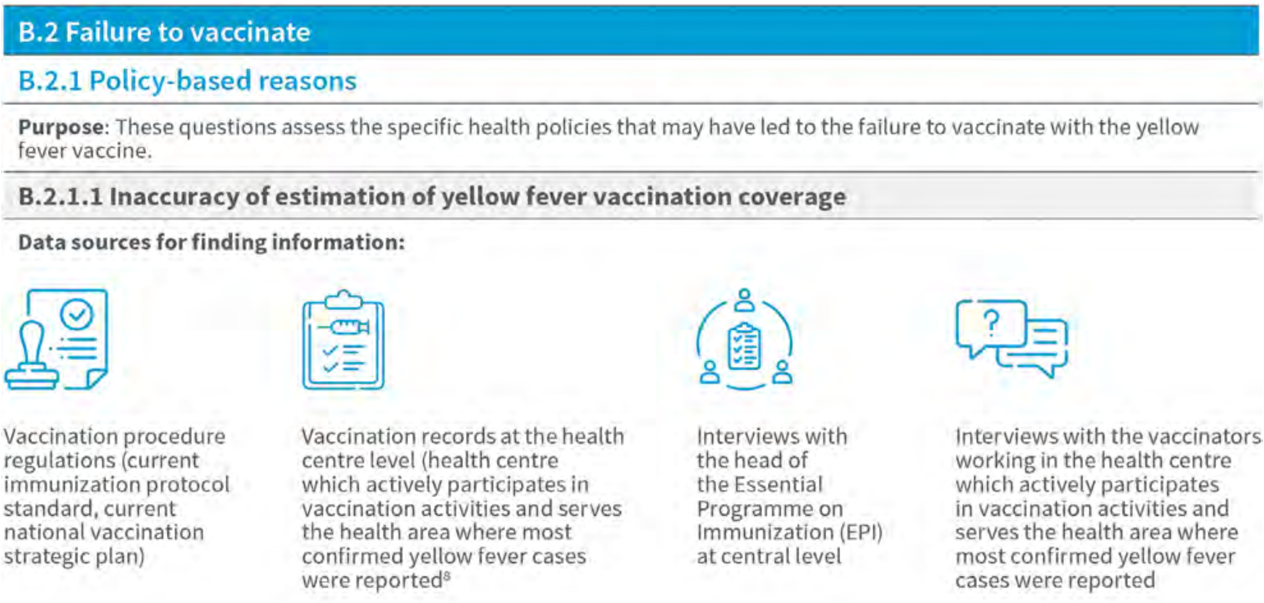
Table 1. Immediate and underlying causes related to a yellow fever outbreak in a district

Failure to vaccinate			
Reasons		**Immediate causes**	**Underlying causes**
Policy-based		• Inaccuracy of estimation of yellow fever vaccination coverage • Gap in the vaccination plan for the general population and population at higher risk for yellow fever	
Provider-based	Routine activities	• Missed routine vaccination sessions • Stock-out of yellow fever vaccines/vaccination equipment • Lack of qualified vaccinators • Lack of knowledge/misconceptions regarding the yellow fever vaccination policy • Lack of knowledge/misconceptions regarding the co-administration of the yellow fever vaccine with another antigen • Lack of knowledge/misconceptions regarding the contraindications of the yellow fever vaccine • Other reasons for not being vaccinating against yellow fever	• Limitation in skills enhancement • Lack of supervision in the vaccination activities
	Mass vaccination campaigns	• Lack of knowledge/misconceptions regarding the co-administration of the yellow fever vaccine with another antigen • Lack of knowledge/misconceptions regarding the contraindications of the yellow fever vaccine • Other reasons for not being vaccinating against yellow fever: ▪ stock-out of vaccines ▪ district not fully covered by the vaccination campaign	• Limitation in skills enhancement • Lack of supervision
User-based		• Vaccines hesitancy/obstacles to vaccination activities (routine and campaigns): ▪ Yellow fever vaccines knowledge and misconceptions ▪ Obstacles to routine vaccination activities ▪ Lack of awareness of the mass vaccination campaigns	
Vaccine failure			
Reasons		**Immediate causes**	**Underlying causes**
Provider-based		• Poor quality of yellow fever vaccines due to: ▪ defective ice-lined refrigerator ▪ poor storage of the yellow fever vaccine and/or diluents during and outside the vaccination sessions • Staff practice failure in the injection	• Lack of trained staff • Misconceptions/lack of knowledge
User-based		• Primary failure • Secondary failure • Host factors	

Sections are clickable for easier navigation

Each subsection that explores a plausible underlying cause for an immunity gap begins by referencing the data sources for finding information, then by recording the information to allow a subsequent analysis.

Box 1. Example of data sources for finding information



B.2 Failure to vaccinate

B.2.1 Policy-based reasons

Purpose: These questions assess the specific health policies that may have led to the failure to vaccinate with the yellow fever vaccine.

B.2.1.1 Inaccuracy of estimation of yellow fever vaccination coverage

Data sources for finding information:

Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)

Vaccination records at the health centre level (health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported[8]

Interviews with the head of the Essential Programme on Immunization (EPI) at central level

Interviews with the vaccinators working in the health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported

The procedures used for data collection are indicated within the questionnaire before each query. Box 2 lists the abbreviations of the procedures.

Box 2. Abbreviations of the procedures for data collection

DR:	document review
EPI C:	interview with the Essential Programme on Immunization (EPI) focal point at central level
EPI D:	interview with the EPI focal point/supervisor at district level
EPI HC:	interview with vaccinators at the health centre level
Dir:	interview with the director of the health centre
Log C:	interview with the logistician of the vaccine storage room at the central level
Log D:	logistics maintenance of the vaccine storage room at the district level
Log HC:	interview with the logistician maintenance of the vaccine storage room at the health centre
CHW:	interview with the community health workers
OBS C:	observation of the vaccine store at central level
OBS D:	observation of the vaccine store at district level
OBS HC:	observation of the vaccine store at health centre level

To evaluate the root causes of immunity gaps, data are collected:

- In the district under investigation and
- In the health centre within that district, which actively participates in vaccination activities and serves the health area where most of the confirmed yellow fever cases were reported.[5]

Data are gathered through either a document review, one-on-one conversation or structured observations.

5 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

Procedures to conduct the document review and observations

Instructions for conducting the document review and observations are provided at each stage of the tool where the review of documents and/or observations are necessary.

Procedures to conduct the one-on-one conversation

The semi-structured questionnaires are conducted with staff involved in vaccination activities. The questionnaires are designed with closed and open-ended questions, allowing flexibility in the interview. The interviewer can probe and follow different directions as information emerges, including flexibility in the sequence of questions. If a response is unclear, the interviewer asks for clarification to ensure to understand the participant's perspective. It is important to minimize notetaking and to consider a voice recorder (with the participant's permission) to capture the conversations.

Note: The text presented in *upper case* letters in the Part B questionnaire serves as an instruction for the interviewer, whereas the text in *lower case* letters are the questions to be asked when conducting review, interviewing participants or conducting observations.

At the end of each subsection that investigates a potential cause of the immunity gap, a summary line allows the user of the questionnaire to review the gathered data and identify if there was a gap on this topic and if 'yes', at which level it occurred.

Box 3. Example of a summary line to review data

EPI HC	23	REVIEW THE IMMUNIZATION RECORDS AT HEALTH CENTRE LEVEL WITH THE VACCINATORS. Are there any discrepancies in the number of children vaccinated between the different data collection sheets and/or between the collection sheets and the document reporting the results to the central level (for example, district health information system 2 (DHIS2), etc.)?	Yes ☐	No ☐	If NO, go to question 25
	24	If YES, please describe the reasons.			
SUMMARY	***25***	***REVIEW THE RESPONSE TO QUESTIONS 13 TO 24.*** ***IS THERE A GAP IN THE ESTIMATION OF YELLOW FEVER VACCINATION COVERAGE:*** • ***IN THE ROUTINE VACCINATION CAMPAIGN?*** • ***IN THE MASS VACCINATION CAMPAIGN?***	 Yes ☐ Yes ☐	 No ☐ No ☐	If NO, go to question 27
	26	***IF YES, AT WHICH LEVEL DOES THE GAP OCCUR?*** • ***CENTRAL LEVEL*** • ***HEALTH CENTRE LEVEL***	 ☐ ☐		

Structure of questionnaire:

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B.2 Failure to vaccinate 29

B.2.1 Policy-based reasons

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B.2.1.2 Gap in the vaccination plan for the general population and populations at higher risk for yellow fever 31

B.2.2 Provider-based reasons

B.2.2.1 Missed routine vaccination sessions 33

B.2.2.2 Stock-out of yellow fever vaccines/vaccination equipment during routine immunization 36

B.2.2.3 Lack of qualified vaccinators for routine immunization 38

B.2.2.4 Lack of knowledge/misconceptions 40

B.2.2.5 Other reasons for not being vaccinated against yellow fever in routine activities 45

B.2.2.6 Other reasons for not being vaccinated against yellow fever during the yellow fever vaccination mass campaign(s) implemented in the district 46

B.2.3 User-based reasons

B.2.3.1 Vaccine hesitancy/obstacles to vaccination activities (routine and campaigns) 47

B.3 Vaccine failure 50

B3.1 Provider-based reasons

B.3.1.1 Suboptimal efficacy of vaccine 50

B.3.1.2 Failure of staff practice in injecting the yellow fever vaccine 57

B.3.2 User-based reasons 57

References 58

QUESTIONNAIRE

Name of the district evaluated: ___________________________________ **Current year:** ______________________

Section A. Yellow fever vaccination coverage estimates in the district under investigation

A.1 Vaccination coverage for routine immunization for the past two years

Purpose: This section reports national and subnational yellow fever vaccination coverage results by data sources.

Data sources for finding information:

- Country profile
- Immunization records on administrative vaccination coverage of yellow fever vaccine through routine immunization
- International Coordinating Group on Vaccine Provision (yellow fever) application form
- WHO/United Nations Children's Fund (UNICEF) estimates of national vaccination coverage data (link)
- Published articles on vaccination
- Weekly Epidemiological Record

1	When was the year of introduction of the yellow fever vaccine in routine immunization?
2	Report routine vaccination coverage over the last 2 years and in the year of the outbreak in the table below.

DR	Level	Years	Source of results (administrative coverage, etc.)	Date when vaccination coverage was calculated (month/year [MM/YYYY])	Vaccination coverage (%)			
					Overall	Girls	Boys	12–23 months
	Central	Year during which the confirmed case(s) were or should have been vaccinated against yellow fever*						
		Year of the outbreak						
		Current year						
		Current year -1						

DR	Level	Years	Source of results (administrative coverage, etc.)	Date when vaccination coverage was calculated (month/year [MM/YYYY])	Vaccination coverage (%)			
					Overall	Girls	Boys	12–23 months
	District	Year during which the confirmed case(s) were or should have been vaccinated against yellow fever*						
		Year of the outbreak						
		Current year						
		Current year -1						
	Health area of the district	Year during which the confirmed case(s) were or should have been vaccinated against yellow fever*						
		Year of the outbreak						
		Current year						
		Current year -1						
		* REFER TO QUESTIONS 8 AND 9 FOR THE YEAR WHEN CASES WERE OR SHOULD HAVE BEEN VACCINATED						

A.2 Vaccination coverage after the mass vaccination campaigns conducted in the district for the past 20 years

Purpose: This section reports vaccination coverage results during yellow fever mass vaccination campaigns if they have taken place in the district.

Data sources for finding information:

- Country profile
- Immunity gap tool
- Reactive vaccination campaign reports
- Preventive vaccination campaign reports
- International Coordinating Group on Vaccine Provision (yellow fever) application form
- WHO/United Nations Children's Fund (UNICEF) estimates of national vaccination coverage data (link)
- Demographic health surveys
- Vaccination articles published
- Weekly Epidemiological Record

DR	3	Were there any mass campaigns of yellow fever in the district?	Yes ☐ No ☐	If NO, go to question 5
	4	If YES, list and describe below the different mass campaigns (reactive [RVC] and/or preventive [PMVC]) that occurred in the district for the past 20 years.[6]		

RVC/ PMVC	Date range of the mass campaigns (MM/YYYY-MM/YYYY)	Age group targeted by the campaigns	Method used to calculate vaccination coverage (administrative/ cluster/other)	Vaccination coverage with recall by individual/ caregiver or with vaccination records?	Vaccination coverage (%)						
					Overall	Women	Men	<1 year	1– <5 years	5– <15 years	≥ 15 years

6 There are different methodologies for measuring vaccination coverage that can be used either to monitor the vaccination campaign (conducted during or after the campaign) or to assess vaccination coverage after the implementation of the campaign. Guidance on the use of these methodologies is presented in chapter 5 of the WHO Eliminate yellow fever epidemics (EYE) strategy country toolkit (2024) *(5)*.

Section B. Reasons contributing to the immunity gaps in the district under investigation

B.1 Generalities

Purpose: These questions assess whether the immunity gap is due to lack of vaccination (= gap in the performance of vaccination program) or vaccine failure (the confirmed case[7] is vaccinated with the yellow fever vaccine).

DR	5	Was an immunity gap analysis conducted in the country?	Yes ☐	No ☐	If NO, go to question 7
	6	If YES, what was the result for the district under investigation?			
	7	Has at least one confirmed case of yellow fever in a recent outbreak been vaccinated with the yellow fever vaccine?	Yes ☐	No ☐	If NO, go to question 9
	8	If YES, indicate the year(s) during which the cases were vaccinated. (IF THE CASES WERE VACCINATED IN DIFFERENT YEARS, NOTE EACH YEAR SEPARATELY.)			
	9	IF ALL OR SOME OF THE CASES WERE NOT VACCINATED, LOOK AT THEIR DATE OF BIRTH AND PROCEED WITH THE QUESTION. OTHERWISE, IF ALL CASES WERE VACCINATED PROCEED TO QUESTION 12. According to the eligibility age of yellow fever vaccination, in which years should unvaccinated cases have been eligible to be vaccinated with the yellow fever vaccine:			
		• through routine immunization			
		• through a mass vaccination campaign.			
	10	GO TO QUESTION 1 AND LOOK AT THE DATE (MM/YYYY) OF THE INTRODUCTION OF THE YELLOW FEVER VACCINE IN THE VACCINATION PROGRAM. Were those unvaccinated cases born after the introduction of the yellow vaccine in the vaccination programmed?	Yes ☐	No ☐	
	11	GO TO QUESTIONS 1 AND 4 AND LOOK AT THE DATES (MM/YYYY) OF THE INTRODUCTION OF THE YELLOW FEVER VACCINE IN THE VACCINATION PROGRAM AND IMPLEMENTATION OF THE MASS CAMPAIGN IF ONE WAS IMPLEMENTED IN THE DISTRICT UNDER INVESTIGATION. Based on the findings, complete the table below.			

Age group (max 10 cases)	Number and percentage of cases that were not vaccinated, but eligible for routine immunization (=born after the introduction of the vaccine in the country).	Number and percentage of cases that were not vaccinated, but eligible for RVCs or PMVCs.	Total number of cases
< 5 years			
5–15 years			
> 15 years			

SUMMARY	12	***ACCORDING TO THE ANSWER TO QUESTION 7, PROCEED WITH DATA COLLECTION ON:*** ***SECTION B.2: SECOND-LEVEL CAUSE: FAILURE TO VACCINATE (QUESTION 13)*** ***OR*** ***SECTION B.3: SECOND-LEVEL CAUSE: VACCINE FAILURE (QUESTION 224)*** ***NOTE: WHEN CONFIRMED CASES ARE NOTED TO BE VACCINATED WITH THE YELLOW FEVER VACCINE BUT PROOF OF SUCH VACCINATION IS MISSING, OR WHEN A PORTION OF THE CONFIRMED CASES HAVE PROOF OF YELLOW FEVER VACCINATION WHILE OTHERS DO NOT, PROCEED WITH DATA COLLECTION FROM BOTH SECTIONS B.2 (QUESTION 13) AND B.3 (QUESTION 224).***

7 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

B.2 Failure to vaccinate

B.2.1 Policy-based reasons

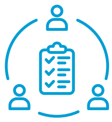
Purpose: These questions assess the specific health policies that may have led to the failure to vaccinate with the yellow fever vaccine.

B.2.1.1 Inaccuracy of estimation of yellow fever vaccination coverage

Data sources for finding information:

Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)

Vaccination records at the health centre level (health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported[8]



Interviews with the head of the Essential Programme on Immunization (EPI) at central level

Interviews with the vaccinators working in the health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported

DR and EPI C	13	Do the denominator estimates for administrative coverage come from: • census data? (if yes, in which year was the census conducted?) • projected population estimates • another source (if yes, specify).	 ☐ ☐ ☐	
	14	Are these data on the immunization target population similar to those from other sources:[9] • regarding routine immunization? • regarding mass vaccination?	 Yes ☐ No ☐ Yes ☐ No ☐	If YES, go to question 16
	15	If NO, why not?		
	16	Is the denominator used for the vaccination coverage accurate: • regarding routine immunization? • regarding mass vaccination?	 Yes ☐ No ☐ Yes ☐ No ☐	
	17	Is the numerator used for the vaccination coverage accurate: • regarding routine immunization? • regarding mass vaccination?	 Yes ☐ No ☐ Yes ☐ No ☐	If YES, go to question 19
	18	If NO, are the reasons: • data entry errors in the tally sheet; • inaccurate recording of the count of vaccinated persons from the tally sheet to the synthesis report at the health centre level; • other (DESCRIBE).	 ☐ ☐ ☐	

8 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

9 The denominators of administrative vaccination coverage are based on population estimates that may be outdated (for example, due to old censuses). It is therefore important to cross-reference data from the target population for routine immunization with other sources of data on births (birth registries and family planning registries) as outlined in the World Health Organization Measles outbreak guide (2022) *(3)*.

	19	LOOK AT QUESTIONS 16 AND 17. IF YOU ANSWER ‘NO’ TO AT LEAST ONE OF THESE QUESTIONS, PROCEED TO THE QUESTION BELOW, OTHERWISE PROCEED TO QUESTION 23. Did any of these gaps lead to an underestimation of the calculation of administrative vaccination coverage of routine or mass activities?	Yes ☐	No ☐	If NO, go to question 21
	20	If YES, please describe.			
	21	Did any of those gaps lead to an overestimation of the calculation of administrative vaccination coverage of routine or mass activities?	Yes ☐	No ☐	If NO, go to question 23
	22	If YES, please describe.			
EPI HC	23	REVIEW THE IMMUNIZATION RECORDS AT HEALTH CENTRE LEVEL WITH THE VACCINATORS. Are there any discrepancies in the number of children vaccinated between the different data collection sheets and/or between the collection sheets and the document reporting the results to the central level (for example, district health information system 2 (DHIS2), etc.)?	Yes ☐	No ☐	If NO, go to question 25
	24	If YES, please describe the reasons.			
SUMMARY	***25***	***REVIEW THE RESPONSE TO QUESTIONS 13 TO 24.*** ***IS THERE A GAP IN THE ESTIMATION OF YELLOW FEVER VACCINATION COVERAGE:*** • ***IN THE ROUTINE VACCINATION CAMPAIGN?*** • ***IN THE MASS VACCINATION CAMPAIGN?***	 Yes ☐ Yes ☐	 No ☐ No ☐	If NO, go to question 27
	26	***IF YES, AT WHICH LEVEL DOES THE GAP OCCUR?*** • ***CENTRAL LEVEL*** • ***HEALTH CENTRE LEVEL***	 ☐ ☐		

B.2.1.2 Gap in the vaccination plan for the general population and populations at higher risk for yellow fever[10]

Data sources for finding information:

Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)

Immunization records at the health centre level (health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported[11]

Interviews with the head of the EPI at central level

Interviews with the district EPI focal point/supervisor

Interviews with the vaccinators working in the health centre which actively participate to vaccination activities and serve the health area where most confirmed yellow fever cases were reported

DR and EPI C	27	What is the age range during which children can receive the yellow fever vaccine as part of the routine activity?			
	28	Are catch-up vaccination sessions available if the child did not get the yellow fever vaccine in this age group?	Yes ☐	No ☐	If NO, go to question 30
	29	If YES, up to what age?			
	30	Do you have populations: • with occupational risk factors? • at higher risk due to sociocultural reasons (for example, marginalized populations, zero dose communities, mobile population)?	 Yes ☐ Yes ☐	 No ☐ No ☐	
	31	Has the latest outbreak impacted on one of these populations?	Yes ☐	No ☐	
	32	Is there a catch-up vaccination strategy for newly-arrived at-risk populations in one district?	Yes ☐	No ☐	If NO, go to question 34
	33	If YES, describe the vaccination strategy (SINCE WHEN, POPULATION AT HIGHER RISK, ETC.).			
	34	Do the children aged (AGE RANGE INDICATED IN QUESTION 27) in the population at higher risk or seasonal workers have access to the country's EPI services?	Yes ☐	No ☐	If NO, go to question 36
	35	If YES, describe.			

10 For the definition of population at higher risk for yellow fever, refer to the WHO Yellow fever: vaccinepreventable diseases surveillance standards manual (2020) *(4)*.

11 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

EPI D	36	Has a catch-up yellow fever vaccination strategy been implemented for a newly-arrived population at higher risk in the district?	Yes ☐	No ☐	If NO, go to question 39
	37	If YES, in which year did this occur?			
	38	If YES, describe the vaccination strategy.			
	39	LOOK AT QUESTION 30 CENTRAL LEVEL RESPONSES. IF AT LEAST ONE “YES”, PROCEED TO THE QUESTION BELOW. IF TWO “NO” ANSWERS, CONTINUE WITH QUESTION 40. Do the children aged (AGE RANGE INDICATED IN QUESTION 27) in the population at higher risk have access to the EPI services?	Yes ☐	No ☐	
EPI HC	40	Has a catch-up yellow fever vaccination strategy been implemented for a newly-arrived population at higher risk in the district?	Yes ☐	No ☐	If NO, go to question 43
	41	If YES, in which year did this occur?			
	42	If YES, describe the vaccination strategy.			
	43	LOOK AT QUESTION 30 CENTRAL LEVEL RESPONSES. IF AT LEAST ONE ‘YES’ PROCEED TO THE QUESTION BELOW, IF TWO ‘NO’ ANSWERS CONTINUE WITH QUESTION 44. Do the children aged (AGE RANGE INDICATED IN QUESTION 27) in the population at higher risk or seasonal workers have access to the EPI?	Yes ☐	No ☐	
SUMMARY	***44***	***REVIEW THE RESPONSES TO QUESTIONS 27 TO 43.*** ***BASED ON THE INFORMATION PROVIDED, ARE THERE ANY GAPS IN THE NATIONAL VACCINATION PLAN THAT MAY HAVE CONTRIBUTED TO MISSED OPPORTUNITIES FOR YELLOW FEVER VACCINATION DURING ROUTINE ACTIVITIES?***	Yes ☐	No ☐	If NO, go to question 46
	45	***IF YES, AT WHICH LEVEL DOES THE GAP OCCUR:*** • ***CENTRAL LEVEL?*** • ***DISTRICT LEVEL?*** • ***HEALTH CENTRE LEVEL?***	 ☐ ☐ ☐		

B.2.2 Provider-based reasons

Purpose: These questions assess the missed opportunities of vaccination that may have led to a failure to vaccinate with the yellow fever vaccine.

B.2.2.1 Missed routine vaccination sessions

Data sources for finding information:

- district vaccination reports
- catch-up vaccination guidance (if available)
- Interviews with the head of the EPI at central level
- Interviews with the district EPI focal point/ supervisor
- Interviews with the director and vaccinators of the health centre which actively participates in vaccination policies and serves the health area where most confirmed yellow fever cases were reported[12]

DR and EPI D

46 GO TO QUESTIONS 1 AND 10. IF THE CASES THAT WERE NOT VACCINATED WERE BORN AFTER THE INTRODUCTION OF THE YELLOW FEVER VACCINE IN THE ROUTINE VACCINATION PROGRAMME, CONTINUE WITH QUESTION 47, OTHERWISE GO TO SECTION B.2.2.4. LACK OF KNOWLEDGE/MISCONCEPTIONS - REGARDING THE CO-ADMINISTRATION OF THE YELLOW FEVER VACCINE WITH ANOTHER ANTIGEN (QUESTION 119).

47 FAMILIARIZE YOURSELF WITH THE DISTRICT AND ITS CATCHMENT AREA ON A MAP. ASK FOR THE NECESSARY REPORTS TO COMPLETE THE TABLE BELOW REGARDING THE VACCINATION SESSIONS (STATIC OR OUTREACH/MOBILE) PLANNED AND COMPLETED.

IF A SESSION DID NOT TAKE PLACE, CALCULATE THE NUMBER OF MISSED SESSIONS AND ASK THE DISTRICT EPI FOCAL POINT WHY THIS HAPPENED.

48 Complete the table

		Health areas affected by the outbreak	**Health areas unaffected by the outbreak**	**Total**
Number of villages in the district	Y YFV*			
	Y*			
Number of children eligible for routine immunization (AGE RANGE INDICATED IN QUESTION 27)	Y YFV			
	Y			
Static sessions				
Number of static sessions planned per month *(a)*	Y YFV			
	Y			
Total number of static sessions planned per year *(b)=a*12*	Y YFV			
	Y			
Total number of static sessions conducted in the year under review *(c)*	Y YFV			
	Y			
*Percentage of planned static sessions conducted in the year under review (c/b * 100)*	Y YFV			
	Y			

12 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccinepreventable diseases surveillance standards manual (2020) *(4)*.

		Outreach and mobile sessions			
		Number of outreach/ mobile sessions planned per month *(d)*	Y YFV		
			Y		
		Total number of outreach/ mobile sessions planned per year *(e)=d*12*	Y YFV		
			Y		
		Total number of outreach/ mobile sessions conducted in the year under review *(f)*	Y YFV		
			Y		
		Percentage of planned outreach/mobile sessions conducted in the year under review *(f/e*100)*	Y YFV		
			Y		
		Total number of sessions			
		Total number of planned sessions per year *(b+e)*	Y YFV		
			Y		
		Total number of sessions conducted per year *(c+f)*	Y YFV		
			Y		
		Percentage of all sessions conducted in the year under review *((c/b*100)+ (f/e*100))*	Y YFV		
			Y		
		*Y: this year; Y YFV: year in which confirmed cases should have been vaccinated with the yellow fever vaccine.			
	49	Give the reasons why the sessions did not take place: (RECORD ALL THAT APPLY.) • lack of staff? ☐ • lack of yellow fever vaccines? ☐ • lack of transportation? ☐ • hard-to-reach area? ☐ • security? ☐ • other (DESCRIBE)? ☐			
EPI HC	50	Have there been any missed vaccination sessions: • the year in which confirmed cases should have been vaccinated? • this year?	Yes ☐ Yes ☐	No ☐ No ☐	IF NO, go to question 55
	51	If YES, were they: • static sessions? ☐ • outreach sessions? ☐ • mobile sessions? ☐			
	52	If YES, how many were missed: • in the year in which the confirmed cases should have been vaccinated? • this year?			
	53	What were the reasons for this: (CHECK ALL THAT APPLY.) • lack of staff? ☐ • lack of yellow fever vaccines? ☐ • lack of transportation? ☐ • hard-to-reach area? ☐ • security? ☐ • other (DESCRIBE)? ☐			
	54	How can the problem be solved?			
SUMMARY	***55***	***REVIEW THE RESPONSES TO QUESTIONS 46 TO 54.*** ***WERE THERE ANY MISSED VACCINATION SESSIONS IN THE DISTRICT IN THE YEAR WHEN THE CONFIRMED CASE(S)[13] WERE SCHEDULED TO BE VACCINATED?***	Yes ☐	No ☐	

13 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

DR and EPI HC	56	REVIEW THE DISTRICT'S EPI VACCINATION SUMMARY REPORT. IF POSSIBLE, ALSO REVIEW THE DATA FROM THE YEAR IN WHICH THE CONFIRMED CASES SHOULD HAVE BEEN VACCINATED.

Complete the table

Year	A	B	C	D	E	F	G	H	I
						Drop-out rate between:			
	Number of registered children	Number of BCG*	Number of Penta 1*	Number of MCV1*	Number of YFV*	Registered children and YFV	BCG and YFV	Penta 1 and YFV	MCV1 and YFV
Year in which the confirmed case(s)** should have been vaccinated -1***									
Current year -1									

*BCG: bacillus Calmette-Guérin; Penta 1: pentavalent first injection; MCV1: measles-containing-vaccine first-dose; YFV: yellow fever vaccine.

** If the confirmed cases were not born in the same year, use the year where most of them were born.

*** Assuming MCV1 schedule is 9–11 months.

57	REVIEW THE RESULTS WITH THE DISTRICT EPI FOCAL POINT AND ASK THE FOLLOWING QUESTIONS IF APPLICABLE. IF NOT APPLICABLE GO TO QUESTIONS 61. Why is the number of children who received the yellow fever vaccine lower than the number of children registered and/or lower than the number of children who received the BCG?
58	How can this problem be addressed?
59	Why is the number of children who received the yellow fever vaccine lower than the number of children who received MCV1?
60	How can this problem be addressed?
61	Report the number of children vaccinated against yellow fever by age and timeliness of vaccination (percentage of invalid and late doses).

Year	Total number of children vaccinated with yellow fever vaccine	Number of vaccinations < 9 months	Number of vaccinations 9–11 months	Number of vaccinations ≥ 12 months	Number of vaccinations <9 months/ number yellow fever (%)	Number of vaccinations 9–11 months/number yellow fever (%)	Number of vaccinations ≥ 12 months/ number yellow fever (%)
Year in which the confirmed case(s)* should have been vaccinated							
This year							

*If the confirmed cases were not born in the same year, use the year when most of them were born.

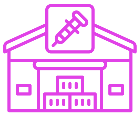
62	ASK IF APPLICABLE THE FOLLOWING QUESTIONS. IF NOT APPLICABLE GO TO QUESTION 64 Why were children vaccinated with yellow fever vaccine after their first birthday?
63	Why were children under 9 months of age vaccinated with the yellow fever vaccine?

SUMMARY	*64*	***REVIEW THE RESPONSES TO QUESTIONS 56 TO 63. WERE THERE ANY GAPS IN COMMUNITY ADHERENCE WITH THE ROUTINE IMMUNIZATION ACTIVITY DURING THE YEAR THE CONFIRMED CASE(S) WERE SCHEDULED TO BE VACCINATED?***

B.2.2.2 Stock-out of yellow fever vaccines/vaccination equipment during routine immunization

Data sources for finding information:

Vaccine books and purchase orders at central and district levels, including the health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported[14]



Interviews with the logistician of the vaccine storage room at central, district and health centre level (health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported)

DR and Log C	65	REVIEW THE STOCK BOOK AND PURCHASE ORDERS WITH THE MANAGER OF THE VACCINE STORAGE ROOM. Have there ever been stock-outs of yellow fever vaccines and/or vaccination equipment at the central level for routine immunization?	Yes ☐	No ☐	If NO, go to question 71
	66	If YES, complete the table below.			
		Item out of stock (for example, yellow fever vaccines, syringes, etc.)	Year (YYYY) of stock shortage	Duration of stock shortage	Reason for stock shortage (for example, supply issues at the supplier level, order delay, poor planning of needs, expired vaccines, lack of funds, etc.)
	67	If YES, has this caused vaccination teams to cancel or shorten planned vaccination sessions?	Yes ☐	No ☐	If NO, go to question 70
	68	If YES, provide details.			
	69	If YES, how can the problem be solved?			
	70	COMPARE THE DATE OF THE STOCK-OUT WITH THE DATE AT WHICH CONFIRMED CASES SHOULD HAVE BEEN VACCINATED (QUESTION 9). Did the two dates occur within a six-month interval?	Yes ☐	No ☐	

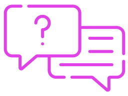
14 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

DR and Log D	71	REVIEW THE YELLOW FEVER VACCINE STOCK BOOK AND PURCHASE ORDERS WITH THE MANAGER OF THE VACCINE STORAGE ROOM. Have there ever been stock-outs of yellow fever vaccines and/or vaccination equipment at the district level for routine immunization?	Yes ☐	No ☐	If NO, go to question 77
	72	If YES, complete the table.			
		Item out of stock (for example, yellow fever vaccines/syringes, etc.) | Year (YYYY) of stock shortage | Duration of stock shortage | Reason for stock shortage (for example, poor planning of needs, expired vaccines, etc.)			
	73	If YES, has this caused vaccination teams to cancel or shorten planned vaccination sessions??	Yes ☐	No ☐	If NO, go to question 76
	74	If YES, provide details.			
	75	If YES, how can the problem be solved?			
	76	COMPARE THE DATE OF THE STOCK-OUT WITH THE DATE AT WHICH CONFIRMED CASES SHOULD HAVE BEEN VACCINATED (QUESTION 9). Did the two dates occur within a six-month interval?	Yes ☐	No ☐	
RD and Log HC	77	Have there ever been stock-outs of yellow fever vaccines and/or vaccination equipment in your health centre for routine immunization?	Yes ☐	No ☐	If NO, go to question 84
	78	If YES, complete the table.			
	79	Item out of stock for example, yellow fever vaccines, syringes, etc.) | Year (YYYY) of stock shortage | Duration of stock shortage | Reason for stock shortage (for example, poor planning of needs, expired vaccines, etc.)			
	80	If YES, has this caused the vaccination team to cancel or shorten planned vaccination sessions?	Yes ☐	No ☐	If NO, go to question 83
	81	If YES, provide details.			
	82	How can the problem be solved?			
	83	COMPARE THE DATE OF THE STOCK-OUT WITH THE DATE AT WHICH CONFIRMED CASES SHOULD HAVE BEEN VACCINATED (QUESTION 9). Did the two dates occur within a six-month interval?	Yes ☐	No ☐	
SUMMARY	***84***	***REVIEW THE RESPONSES TO QUESTIONS 65 TO 83.*** ***WAS THERE A YELLOW FEVER VACCINE STOCK-OUT THAT LED TO THE CANCELLATION OR THE SHORTENING OF ROUTINE VACCINATION SESSIONS IN THE YEAR WHEN CONFIRMED CASE (S)[15] SHOULD HAVE BEEN VACCINATED?***	Yes ☐	No ☐	If NO, go to question 86
	85	***IF YES, AT WHICH LEVEL DID THE STOCK-OUT OCCUR?*** • ***CENTRAL LEVEL?*** ☐ • ***DISTRICT LEVEL?*** ☐ • ***HEALTH CENTRE LEVEL?*** ☐			

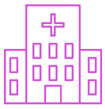
15 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020)*(4)*.

B.2.2.3 Lack of qualified vaccinators for routine immunization

Data sources for finding information:



Interviews with the district EPI focal point/supervisor



Interviews with the director and vaccinators working at the health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported

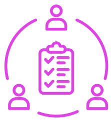
EPI D	86	How many health centres offer vaccination sessions in the district this year?			
	87	How many sanctioned vaccinators are there in the district this year?			
	88	How many sanctioned vaccinators are there on average per health centre offering vaccination sessions in the district this year? (=QUESTIONS 87/86)			
	89	How many children aged (AGE RANGE INDICATED IN QUESTION 27) per one sanctioned vaccinator are there on average in the district? (=questions 48/87)			
	90	Does the number of sanctioned vaccinators meet the recommended standard?	Yes ☐	No ☐	
	91	From a supervisor's perspective, do you think that the number of sanctioned vaccinators working in the district is sufficient to conduct all planned vaccination sessions?	Yes ☐	No ☐	If YES, go to question 93
	92	If NO, what are the reasons why the number of sanctioned vaccinators in the district is not sufficient: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT SUPERVISORS CONSIDER RELEVANT.) • no funds for recruitment? ☐ • not enough training? ☐ • other (DESCRIBE)? ☐			
	93	Has the lack of sanctioned vaccinators prevented the vaccination team from organizing routine vaccination activities in the district?	Yes ☐	No ☐	If NO, go to question 95
	94	IF YES, COMPARE THE DATE AT WHICH THERE WAS A SHORTAGE OF VACCINATORS WITH THE DATE AT WHICH CONFIRMED CASES SHOULD HAVE BEEN VACCINATED (QUESTION 9). Did the two dates occur within a six-month interval?	Yes ☐	No ☐	

EPI HC	95	How many positions of sanctioned vaccinators are there working in the health centre to cover your health area?			
	96	Are there other positions for vaccinators (volunteers) in your health centre?	Yes ☐	No ☐	
	97	How many vaccinators do you have in total?			
	98	Do you think that this number of vaccinators is sufficient to conduct all the vaccination sessions planned in your health area?	Yes ☐	No ☐	If YES, go to question 101
	99	If NO, explain why.			
	100	If NO, why are there not more vaccinators: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT THE IMMUNIZATION PROVIDERS CONSIDER RELEVANT) no funds for recruitment? ☐ not enough training? ☐ other (DESCRIBE)? ☐			
	101	Has a lack of vaccinators ever prevented the vaccination team from conducting routine vaccination sessions?	Yes ☐	No ☐	If NO, go to question 104
	102	If YES, which year?			
	103	IF YES, COMPARE THE YEAR IN WHICH THERE WAS A LACK OF VACCINATORS WITH THE YEAR IN WHICH THE CONFIRMED CASES SHOULD HAVE BEEN VACCINATED (QUESTION 9). Did the two dates occur within a six-month interval?	Yes ☐	No ☐	
SUMMARY	***104***	***REVIEW THE RESPONSES TO QUESTIONS 86 TO 103.***			
	105	***WAS THERE A LACK OF VACCINATORS THAT LED TO THE CANCELLATION OF ROUTINE VACCINATION SESSIONS IN THE YEAR IN WHICH CONFIRMED CASES SHOULD HAVE BEEN VACCINATED?***	Yes ☐	No ☐	

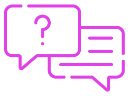
B.2.2.4 Lack of knowledge/misconceptions

Data sources for finding information:

Vaccination procedure regulations (current immunization protocol standard, current national vaccination strategic plan)



Interviews with the head of the EPI at central level



Interviews with the district EPI focal point/ supervisor

Interviews with the director and vaccinators working at the health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported[16]

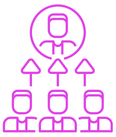
a. Regarding yellow fever vaccination policy					
Failure to open vials if a certain number of children are not present at the routine vaccination session					
DR and EPIC	106	For how many children can a vial of yellow fever vaccine be opened during a vaccination session?			
EPI D	107	How many children have to be present before the vaccinator opens a yellow fever vaccine vial? (CHECK IF THE ANSWER IS IN LINE WITH THE OPENING POLICY QUESTION 106.)			
EPI HC	108	Do you open a vial of yellow fever vaccine if there is only one child present at the vaccination session? (CHECK IF THE ANSWER IS IN LINE WITH THE OPENING POLICY QUESTION 106.)	Yes ☐	No ☐	If YES, go to question 111
	109	If NO, how many children must be present before a yellow fever vaccine vial is opened?			
	110	If NO, what do you say to the caregiver of the child who came to be vaccinated for yellow fever and is the only child present at the vaccination session?			
Vaccination schedule of the routine immunization activity					
	111	If a caregiver brings a child over (AGE RANGE INDICATED IN QUESTION 27) months to be vaccinated against yellow fever, should the vaccinator administer the vaccine?	Yes ☐	No ☐	If NO, go to question 113
	112	If YES, up to what age can the vaccinator administer the yellow fever vaccine? (TO BE CHECKED WITH QUESTION 29 WHEN THE COUNTRY CONDUCT CATCH-UP VACCINATION SESSION.)			Go to question 114
	113	If NO, what are the reasons: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT THE SUPERVISOR CONSIDERS RELEVANT.) • the vaccination schedule does not authorize the administration of yellow fever vaccine after 11 months of age? ☐ • this creates problems for recording data in the vaccination file? ☐ • administration of yellow fever vaccine after 11 months of age leads to vaccine failure? ☐ • I consider that vaccinating children >11 months of age to be a waste of resources? ☐ • other (DESCRIBE)? ☐			

16 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

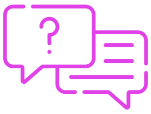
EPI HC	114	If a caregiver brings a child over (AGE RANGE INDICATED IN QUESTION 27) months of age to be vaccinated against yellow fever, do you administer the vaccine?	Yes ☐	No ☐	If NO, go to question 116
	115	If YES, up to what age do you administer yellow fever vaccine? (TO BE CHECKED WITH QUESTION 29 WHEN THE COUNTRY CONDUCT CATCH-UP VACCINATION SESSION.)			Go to question 117
	116	If NO, what are the reasons: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT THE VACCINATOR CONSIDERS RELEVANT.) • the vaccination schedule does not allow the administration of yellow fever vaccine after 11 months of age? ☐ • this creates problems for recording data in the vaccination file? ☐ • administration of the yellow fever vaccine after 11 months of age results in vaccine failure? ☐ • I consider vaccinating children >11 months of age to be a waste of resources? ☐ • other (DESCRIBE)? ☐			
Access to the routine immunization service					
DR and EPI C	117	According to the country vaccination policy, can children eligible for the routine EPI be vaccinated against yellow fever:			
		in all health centres in the country?	☐		
		in specific health centres according to their area of residence?	☐		
EPI HC	118	What do you do if a mother comes with her child to be vaccinated against yellow fever during a routine vaccination session, even though they do not reside in the health area? (TO BE CHECKED WITH QUESTION 117.)			
b. Regarding the co-administration of the yellow fever vaccine with another antigen (during routine activity or mass vaccination campaigns)					
DR and EPI C	119	Is there a policy of co-administration of yellow fever vaccine with other antigens (for example, measles) during routine activities or mass vaccination campaigns?	Yes ☐	No ☐	
EPI D	120	Is it possible to co-administer yellow fever vaccine at the same time as another vaccine? (TO BE CHECKED WITH QUESTION 119.)	Yes ☐	No ☐	
EPI HC	121	Is it possible to administer the yellow fever vaccine at the same time as another vaccine? (TO BE CHECKED WITH QUESTION 119.)	Yes ☐	No ☐	If YES, go to question 123
	122	If NO, why not: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT THE VACCINATOR CONSIDERS RELEVANT.) • risk of interaction between the two vaccines? ☐ • this is not part of the vaccine policy? ☐ • reduction in the effectiveness of vaccines? ☐ • dangerousness? ☐ • other (DESCRIBE)? ☐			

c. Regarding the contraindications of the yellow fever vaccine (during routine activities and mass vaccination campaigns)					
EPI HC	123	Can a child receive the yellow fever vaccine if the child has:			
		• a mild respiratory illness (for example, cough, runny nose)?	☐		
		• diarrhoea?	☐		
		• proven recent contact with a sick person?	☐		
		• experienced a previous local reaction from vaccination (see list below):	☐		
		▪ redness	☐		
		▪ swelling	☐		
		▪ low fever	☐		
		▪ crying	☐		
		▪ soreness, pain	☐		
		▪ only if two or more of the above?	☐		
		• egg allergy?	☐		
		• previous anaphylactic or severe allergic reaction to vaccine?	☐		
		• severe illness (for example, high fever)?	☐		
		• child is breastfeeding?	☐		
SUMMARY	***124***	***REVIEW THE RESPONSES TO QUESTIONS 106 TO 123 (IF YOU STARTED AT QUESTION 119, REVIEW THE RESPONSES TO QUESTIONS 119 TO 123).*** ***WAS THERE A LACK OF KNOWLEDGE OR MISCONCEPTION ABOUT THE VACCINATION POLICY AND/OR THE YELLOW FEVER VACCINE AMONG VACCINATORS THAT RESULTED IN A MISSED OPPORTUNITY FOR YELLOW FEVER VACCINATION?***	Yes ☐	No ☐	***If NO, go to question 160***
	125	***IF YES, WAS THE MISCONCEPTION RELATED TO:***			
		• THE VACCINATION POLICY?	☐		
		• THE YELLOW FEVER VACCINE?	☐		
	126	***IF YES, AT WHICH LEVEL DID THE GAP OCCUR:***			
		• CENTRAL LEVEL?	☐		
		• DISTRICT LEVEL?	☐		
		• HEALTH CENTRE LEVEL?	☐		

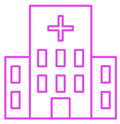
Data sources for finding information:



Record of supervisory visits



Interviews with the district EPI focal point/supervisor



Interviews with the director and vaccinators working at the health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported[17]

Limitation in skills enhancement					
EPI D	127	Are there any vaccine policy guidelines available for vaccinators?	Yes ☐	No ☐	If NO, go to question 129
	128	If YES, since when?			
	129	What types of training do vaccinators receive?			
	130	How often are they trained: • annually? • quarterly? • ad hoc? • other (DESCRIBE)?	 ☐ ☐ ☐ ☐		
	131	Do you think that vaccinators working in district health centres need additional training on immunization to do their work properly?	Yes ☐	No ☐	If NO, go to question 133
	132	If YES, what training(s) would be required?			
	133	Did the absence of this training(s) lead to any specific problems?	Yes ☐	No ☐	If NO, go to question 136
	134	If YES, describe the problems this has caused.			
	135	If YES, how can the problem be solved?			
EPI HC	136	As a vaccinator, would you need additional training in immunization to perform your duties?	Yes ☐	No ☐	If NO, go to question 138
	137	If YES, what training would be required?			
	138	Did the absence of this training lead to any specific problems?	Yes ☐	No ☐	If NO, go to question 141
	139	If YES, describe the problems this has caused?			
	140	If YES, how can the problem be solved?			

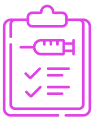
17 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

Lack of supervision in vaccination activities					
EPI D	141	How many sanctioned supervisor positions were there in the district: • the year in which confirmed cases should have been vaccinated? • this year?			
	142	How many of them have been filled: • the year in which confirmed cases should have been vaccinated? • this year?			
	143	How many health centres offering vaccination sessions are there per supervisor (= QUESTIONS 86/142)?			
	144	Do you think that the number of sanctioned supervisors working in the district is enough to complete all planned supervision sessions?	Yes ☐	No ☐	If YES, go to question 146
	145	If NO, what are the reasons why the number of sanctioned supervisors in the district is not sufficient: (LIST THE FOLLOWING ANSWERS. TICK ALL THE ANSWERS THAT THE SUPERVISOR CONSIDERS RELEVANT.) • no funds for recruitment? ☐ • not enough training? ☐ • other (DESCRIBE)? ☐			
	146	How many supervision visits for routine immunization activities were planned in all district health centres in the year yellow fever cases were reported?			
	147	How many of these supervision visits were conducted?			
	148	How many routine immunization supervision visits were planned at the health centre(s) affected by the yellow fever outbreak?			
	149	How many of these supervision visits were conducted?			
	150	Has there been any problems that made it difficult to supervise routine immunization activities in health centres offering vaccination sessions?	Yes ☐	No ☐	If NO, go to question 154
	151	If YES, what were they: • transport unavailable? ☐ • limited budget? ☐ • hard-to-reach areas (security, geographical barriers, etc.)? ☐ • competing priorities (for example, Covid-19-related activities)? ☐ • other (DESCRIBE)? ☐			
	152	If YES, how can this problem be solved?			
	153	If YES, did this occur in the health centre where the confirmed cases should have been vaccinated through the routine immunization activity?	Yes ☐	No ☐	
EPI HC	154	How many times has the district supervisor visited your health centre: • during the year in which confirmed case(s) should have been vaccinated? • this year?			
	155	Do you think that the supervision of vaccinators and vaccination sessions is adequate?	Yes ☐	No ☐	If YES, go to question 158
	156	If NO, why is it inadequate?			
	157	If NO, what can be done to improve the supervision?			

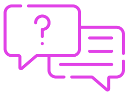
SUMMARY	***158***	***REVIEW THE RESPONSES TO QUESTIONS 127 TO 157*** ***WAS THE KNOWLEDGE GAP IDENTIFIED IN QUESTION 124 ATTRIBUTABLE TO:*** • ***LIMITATIONS IN SKILLS DEVELOPMENT?*** ☐ • ***INSUFFICIENT SUPERVISION IN VACCINATION ACTIVITIES?*** ☐			
	159	***AT WHICH LEVEL DOES THE GAP OCCUR:*** • ***CENTRAL LEVEL?*** ☐ • ***DISTRICT LEVEL?*** ☐ • ***HEALTH CENTRE LEVEL?*** ☐			

B.2.2.5 Other reasons for not being vaccinated against yellow fever in routine activities

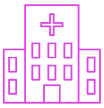
Data sources for finding information:



District vaccination reports



Interviews with the district EPI focal point/supervisor



Interviews with the director and vaccinators working in the health centre in which actively participates in vaccination activities and serves the health area where most confirmed cases were reported[18]

EPI C	160	GO TO QUESTION 10. IF THE CONFIRMED CASES THAT WERE NOT VACCINATED WERE BORN AFTER THE INTRODUCTION OF THE YELLOW FEVER VACCINE IN THE ROUTINE VACCINATION PROGRAMME, CONTINUE WITH THE QUESTION BELOW; OTHERWISE GO TO SECTION B.2.2.6 OTHER REASONS FOR NOT BEING VACCINATED AGAINST YELLOW FEVER DURING THE YELLOW FEVER VACCINATION MASS CAMPAIGN(S) IMPLEMENTED IN THE DISTRICT (QUESTION 169). Are there any other issues you can think of that have cancelled routine vaccination activities in the year in which the cases should have been vaccinated? (REFER TO QUESTION 9 TO FIND THE YEAR IN WHICH THE CASES SHOULD HAVE BEEN VACCINATED.)	Yes ☐	No ☐	If NO, go to question 162
	161	If YES, which ones?			
EPI D	162	Are there any other issues you can think of that have cancelled vaccination activities in the year in which the cases should have been vaccinated? (REFER TO QUESTION 9 TO FIND THE YEAR IN WHICH THE CASES SHOULD HAVE BEEB VACCINATED).	Yes ☐	No ☐	If NO, go to question 164
	163	If YES, which ones?			
EPI HC	164	Are there any other issues you can think of that have cancelled vaccination activities in the year in which the cases should have been vaccinated? (REFER TO QUESTION 8 TO FIND THE YEAR IN WHICH THE CASES SHOULD HAVE BEEB VACCINATED.)	Yes ☐	No ☐	If NO, go to question 167
	165	If YES, which ones?			
	166	If YES, how can this problem be solved?			
SUMMARY	***167***	***REVIEW THE RESPONSES TO QUESTIONS 160 TO 166.*** ***WERE THERE ADDITIONAL FACTORS THAT HAVE CONTRIBUTED TO VACCINATION MISSED OPPORTUNITIES DURING ROUTINE SESSIONS?***	Yes ☐	No ☐	If NO, go to question 169
	168	***IF YES, DESCRIBE WHICH ONE(S) AND AT WHICH LEVEL THE GAP OCCURRED:*** • ***CENTRAL LEVEL?*** ☐ • ***DISTRICT LEVEL?*** ☐ • ***HEALTH CENTRE LEVEL?*** ☐			

18 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

B.2.2.6 Other reasons for not being vaccinated against yellow fever during the yellow fever vaccination mass campaign(s) implemented in the district

Data sources for finding information:

Interviews with the district EPI focal point/supervisor

DR	169	Was there one or several yellow fever vaccination campaigns implemented in the district (REFER TO QUESTION 4)?	Yes ☐	No ☐	If NO, go to question 177
	170	If YES, were the confirmed case(s) eligible for the mass vaccination campaign?	Yes ☐	No ☐	If NO, go to question 177
EPI D	171	Were there any areas in the district that were not reached by the vaccination campaign?	Yes ☐	No ☐	If NO, go to question 173
	172	If YES, what were the causes: (TICK ALL THE ANSWERS THAT THE SUPERVISOR CONSIDERS RELEVANT.) • insufficient number of vaccination teams? • difficulty to access certain areas? • security issues? • lack of microplanning? • vaccine stock-out during the campaign? • cold chain issues • other (DESCRIBE)?	 ☐ ☐ ☐ ☐ ☐ ☐ ☐		
	173	Was a communication and social mobilization plan for the mass campaign implemented in the district?	Yes ☐	No ☐	If NO, go to question 175
	174	If NO, explain the reasons.			
SUMMARY	***175***	***REVIEW THE RESPONSES TO QUESTIONS 169 TO 174. WERE THERE ADDITIONAL FACTORS THAT HAVE CONTRIBUTED TO VACCINATION MISSED OPPORTUNITIES DURING THE MASS VACCINATION CAMPAIGNS?***	Yes ☐	No ☐	If NO, go to question 177
	176	***IF YES, DESCRIBE WHICH ONE(S) AND AT WHICH LEVEL THE GAP OCCURRED: • CENTRAL LEVEL? ☐ • DISTRICT LEVEL? ☐ • HEALTH CENTRE LEVEL? ☐***			

B.2.3 User-based reasons

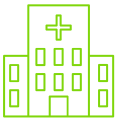
Purpose: These questions assess the user-based reasons that may have led to a failure to vaccinate with the yellow fever vaccine.

B.2.3.1 Vaccine hesitancy/obstacles to vaccination activities (routine and campaigns)

Data sources for finding information:

Published articles on vaccination acceptance

Interviews with the district EPI focal point/ supervisor



Interviews with the director and vaccinators working at the health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported[19]

Interviews with community health workers

Interviews with the confirmed case(s) or the caregiver of the confirmed case(s)

DR	177	Has there been any data collected/evidence gathered on vaccine acceptance, particularly for yellow fever vaccine?	Yes ☐	No ☐	If NO, go to question 179
	178	If YES, describe the results.			
EPI D	179	Are vaccines generally well accepted in the district?	Yes ☐	No ☐	
	180	Is acceptance of the yellow fever vaccine similar to other vaccines?	Yes ☐	No ☐	If YES, go to question 183
	181	If NO, why not?			
	182	If NO, how can this problem be solved?			
	183	Are you aware of any rumours, misconceptions, myths and/ or beliefs (current or past) about vaccines and vaccination in general in the district?	Yes ☐	No ☐	If NO, go to question 186
	184	IF YES, please indicate the rumours, myths, beliefs and/or misconceptions.			
	185	If YES, when was it? (RECORD THE TIME FRAME IF THE PERSON DOES NOT REMEMBER THE EXACT YEAR, SUCH AS BEFORE, DURING OR AFTER THE COVID-19 PANDEMIC, ETC.)			
	186	Does the year coincide with the one in which the cases were expected to be vaccinated, either through routine or mass vaccination campaigns (REFER TO QUESTION 9)?	Yes ☐	No ☐	If NO, go to question 188
	187	If YES, how can these rumours, myths, beliefs and/or misconceptions be addressed?			
	188	How did the Covid-19 pandemic impact on vaccination activities?			
	189	In your opinion, what obstacles could prevent EPI users from coming to health centres to be vaccinated?			
	190	How can this problem be solved?			

19 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

EPI HC	191	Are vaccines generally well accepted in the health area?	Yes ☐	No ☐	
	192	Is acceptance of the yellow fever vaccine similar to other vaccines?	Yes ☐	No ☐	If YES, go to question 195
	193	If NO, why not?			
	194	If NO, how can this problem be solved?			
	195	Are you aware of any rumours, misconceptions, myths and/or beliefs (current or past) about vaccines and vaccination in general in the health area?	Yes ☐	No ☐	If NO, go to question 198
	196	If YES, please describe these rumours, myths, beliefs and misconceptions.			
	197	If YES, when was it? (RECORD THE TIME FRAME IF THE PERSON DOES NOT REMEMBER THE EXACT YEAR, SUCH AS BEFORE, DURING OR AFTER THE COVID-19 PANDEMIC, ETC.)			
	198	Does the year coincide with the one in which the cases were expected to be vaccinated, either through routine or mass vaccination campaigns (REFER TO QUESTION 9)?	Yes ☐	No ☐	
	199	How did the Covid-19 pandemic impact vaccinations activities?			
	200	In your opinion, what obstacles could prevent EPI users from coming to health centres to be vaccinated during a routine session?			
	201	How can this problem be solved?			
	202	Are there awareness activities for routine immunization activities in your health area?	Yes ☐	No ☐	If NO go to question 204
	203	If YES, how often?			
	204	How many community health workers cover the health area for routine immunization?			
	205	How many outreach sessions regarding vaccine awareness for routine immunization do you have in the health area per month?			
	206	Were there any events in the past that prevented the organization of awareness sessions?	Yes ☐	No ☐	If NO, go to question 208
	207	If YES, which ones?			
CHW	208	How accepted are yellow fever vaccines in the health area: • low? ☐ • moderate? ☐ • high? ☐			
	209	Is acceptance of the yellow fever vaccine similar to other vaccines?	Yes ☐	No ☐	If YES, go to question 212
	210	If NO, why not?			
	211	If NO, how can this problem be solved?			
	212	Are you aware of any rumours, misconceptions, myths and/or beliefs (current or past) about vaccines and vaccination in general in the health area?	Yes ☐	No ☐	If NO, go to question 215
	213	If YES, please indicate the rumours, myths, beliefs and/or misconceptions?			
	214	If YES, do you remember when it was? (RECORD THE TIME FRAME IF THE PERSON DOES NOT REMEMBER THE EXACT YEAR, SUCH AS BEFORE, DURING OR AFTER THE COVID-19 PANDEMIC, ETC.)			
	215	Does the year coincide with the one in which the cases were expected to be vaccinated, either through routine or mass vaccination campaigns (REFER TO QUESTION 9)?	Yes ☐	No ☐	If NO, go to question 217
	216	In your opinion, what obstacles could prevent EPI users from coming to health centres to be vaccinated during the routine sessions?			

Caregiver of the confirmed case(s)	217	**GO TO QUESTION 10 TO ANSWER THE FOLLOWING QUESTION** Was the yellow fever case born after the introduction of the yellow fever vaccine in the national vaccination schedule (routine program)?	Yes ☐	No ☐	If NO, go to question 219
	218	IF YES, ASK THE CAREGIVERS OF THE CONFIRMED CASES (UP TO THREE SETS OF CAREGIVERS IF MULTIPLE CASES). IF THE CAREGIVERS ARE NOT FOUND, GO TO QUESTION 219. Why did you not bring your child to the routine immunization session to get the yellow fever vaccine: • competing activities at home or outside when the vaccination activities took place? • there is/was a long queue on the routine vaccination site? • I am/was not allowed to leave home? • I do/did not trust the vaccines? • I have/had religious objections to vaccination? • It is/was too costly to go to a vaccination session (CONSIDER ANY PAYMENTS TO THE CLINIC, THE COST OF GETTING THERE, PLUS THE COST OF TAKING TIME OFF WORK AND TO TREAT EVENTUAL SIDE-EFFECTS)? • I am/was concerned that the yellow fever vaccine may cause serious, life-threatening adverse reactions or death? If so, which ones? • I am/was concerned that yellow fever vaccine may cause minor reactions, such as pain, swelling, redness and fever? If so, which ones? • I am/was concerned that my child is already getting too many injections of different vaccines? • the yellow fever vaccine is/was not available at the site? • the vaccinator is/was not available at the site? • the vaccinator cannot/could not speak or understand the language of the caregivers? • the vaccinator is/was not friendly? • the vaccinator did not give the yellow fever vaccine because the child was ill? • the vaccination site location is/was not suitable? • the timing of vaccination session is/was not suitable? • other (DESCRIBE)?			
Confirmed case(s) or their caregiver if children	219	Was there one or several yellow fever vaccination campaigns implemented in the district (REFER TO QUESTION 4)?	Yes ☐	No ☐	If NO, go to question 223
	220	If YES, were the confirmed case(s) eligible for the mass vaccination campaign?	Yes ☐	No ☐	If NO, go to question 223
	221	Where were you in the year of the mass vaccination campaign?			
	222	What were the reasons why you were not vaccinated during the mass vaccination campaign that occurred in (REFER TO QUESTION 4 TO FIND THE DATE OF THE MASS VACCINATION CAMPAIGN) in the district: • I had competing activities at home or outside when the vaccination campaign took place? • there was a long queue in the vaccination site? • it is/was too costly to go to a vaccination session (consider the cost of getting to the vaccination site, plus the cost of taking time off work, and to treat eventual side-effects)? • other (DESCRIBE)?			
SUMMARY	***223***	***REVIEW THE RESPONSES TO QUESTIONS 177 TO 222.*** ***FROM THE USER'S PERSPECTIVE, IDENTIFY THE REASONS THAT PREVENTED THE CONFIRMED CASES FROM RECEIVING THE YELLOW FEVER VACCINE, WHETHER THROUGH ROUTINE ACTIVITIES OR MASS VACCINATION CAMPAIGNS, IF SUCH A CAMPAIGN WAS CONDUCTED IN THE DISTRICT.***			

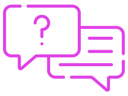
B.3 Vaccine failure

B3.1 Provider-based reasons

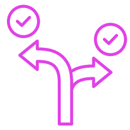
B.3.1.1 Suboptimal efficacy of vaccine

Factor related to defective ice-lined refrigerator (IRL)

Data sources for finding information:



Interview with the logistics managers of the vaccine storage room at the central, district and health centre level (which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported);



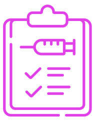
Contingency plan

Log C	224	Is there a contingency plan at the central level for dealing with defective IRLs?	Yes ☐	No ☐	If NO, go to question 226
	225	If YES, ask for a copy and since when?			
	226	Has one of the ILRs containing yellow fever vaccine at central level been faulty at any time?	Yes ☐	No ☐	If NO, go to question 228
	227	IF YES, COMPARE THE DATE WHEN THE IRL WAS REPORTED DEFECTIVE WITH THE DATE THE CONFIRMED CASES WERE VACCINATED (QUESTION 8) AND PROCEED WITH THE QUESTION. Did the two dates occur within a six-month interval?	Yes ☐	No ☐	
	228	What procedures were followed for the yellow fever vaccine during this period?			
Log D	229	Is there a contingency plan on cold chain and vaccine management at the district level?	Yes ☐	No ☐	If NO, go to question 231
	230	If YES, ask for a copy and since when?			
	231	What is the type of power supply for the IRL: • electricity as the main source? ☐ • solar energy as the main source? ☐ • kerosene as the main source? ☐ • other (DESCRIBE)? ☐			
	232	Has the ILR been faulty at any point?	Yes ☐	No ☐	If NO, go to question 236
	233	If YES, on what date was the ILR maintained?			
	234	IF YES, COMPARE THE DATE THE IRL WAS REPORTED DEFECTIVE WITH THE DATE THE CONFIRMED CASES WERE VACCINATED (QUESTION 8) AND PROCEED WITH THE QUESTION. Did the two dates occur within a six-month interval?	Yes ☐	No ☐	
	235	What procedures were followed for the yellow fever vaccine during this period?			

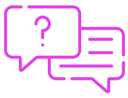
Log HC	236	Is there a contingency plan on cold chain and vaccine management at the district level?	Yes ☐	No ☐	If NO, go to question 238
	237	If YES, ask for a copy and since when?			
	238	What is the type of power supply for the refrigerator/IRL: • electricity as the main source? ☐ • solar energy as the main source? ☐ • kerosene as the main source? ☐ • other (DESCRIBE)? ☐			
	239	Has the ILR been faulty at any point?	Yes ☐	No ☐	If NO, go to question 243
	240	If YES, on what date was the ILR maintained?			
	241	IF YES, COMPARE THE DATE THE IRL WAS REPORTED DEFECTIVE WITH THE DATE THE CONFIRMED CASES WERE VACCINATED (QUESTION 8) AND PROCEED WITH THE QUESTION. Did the two dates occur within a six-month interval?	Yes ☐	No ☐	
	242	What procedures were followed for the yellow fever vaccine during this period?			

Factor related to poor storage of the yellow fever vaccine and/or the diluents during and outside the vaccination sessions

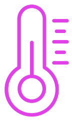
Data sources for finding information:



Observation of the vaccine's storage room at central, district and health centre levels (health centre which actively participates in vaccination policies and serves the health area where most confirmed yellow fever cases were reported)



Interview with the logistics managers of the vaccine storage room at the central, district and health centre level (health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported)[20]



temperature monitoring sheet

OBS C	243	ASK THE PERSON IN CHARGE OF THE VACCINE STORE IF IT IS POSSIBLE TO LOOK AT THE PLACE WHERE THE VACCINES, ICE PACKS AND DILUENTS ARE STORED. FILL OUT THE QUESTIONNAIRES BELOW BASED ON THE OBSERVATIONS MADE.			
		Are the yellow fever vaccines stored properly in the IRL?	Yes ☐	No ☐	
	244	Are there any yellow fever vaccines with the vaccine vial monitor at stage 3 or 4?	Yes ☐	No ☐	If NO, go to question 246
	245	If YES, how many vials?			
	246	Are there any expired yellow fever vaccine vials?	Yes ☐	No ☐	If NO, go to question 248
	247	If YES, how many vials?			
	248	Are the ice packs in the freezer: • completely frozen? • partially frozen? • not frozen?			
	249	Are the ice packs filled to the correct level (no leaks)?	Yes ☐	No ☐	

20 For the current yellow fever confirmed case definition, please refer to the WHO Yellow fever: vaccine-preventable diseases surveillance standards manual (2020) *(4)*.

	250	Are they positioned correctly in the freezer?	Yes ☐	No ☐	
	251	Is there a functioning temperature monitoring device in the freezer?	Yes ☐	No ☐	If NO, go to question 253
	252	If YES, what is the temperature of the freezer?			
	253	Is there a monitoring chart on the freezer?	Yes ☐	No ☐	If NO, go to question 255
	254	If YES, since when?			
	255	Is there an IRL temperature monitoring device?	Yes ☐	No ☐	If NO, go to question 257
	256	If YES, what is the temperature inside the IRL?			
	257	Is there a monitoring chart on the IRL?	Yes ☐	No ☐	If NO, go to question 259
	258	If YES, how many times a day is the temperature recorded on the monitoring sheet?			
	259	Was there a failure in the temperature recording this year?	Yes ☐	No ☐	If NO, go to question 261
	260	If YES, when?			
	261	Have temperatures been outside the 2–8°C range this year (lower or higher)?	Yes ☐	No ☐	If NO, go to question 263
	262	If YES, when and what was the temperature?			
	263	Are there monitoring charts for the year of vaccination of confirmed cases (REFER TO QUESTION 8)?	Yes ☐	No ☐	If NO, go to question 266
	264	If YES, was there a failure in the temperature record that year?	Yes ☐	No ☐	If NO, go to question 266
	265	If YES, when?			
	266	Were temperatures outside the 2–8°C range that year (lower or higher)?	Yes ☐	No ☐	If NO, go to question 268
	267	If YES, when and what was the temperature?			
	268	Are the solvents kept at the right temperature according to the vaccine manufacturer?	Yes ☐	No ☐	
OBS D	270	ASK THE PERSON IN CHARGE OF THE VACCINE STORE IF IT IS POSSIBLE TO LOOK AT THE PLACE WHERE THE VACCINES ARE STORED. FILL OUT THE QUESTIONNAIRES BELOW BASED ON THE OBSERVATIONS MADE. Are the yellow fever vaccines kept stored properly in the IRL?	Yes ☐	No ☐	
	271	Are there any yellow fever vaccine with the vaccine vial monitors at stage 3 or 4?	Yes ☐	No ☐	If NO, go to question 273
	272	If YES, how many vials?			
	273	Are there any expired yellow fever vaccine vials?	Yes ☐	No ☐	If NO, go to question 275
	274	If YES, how many vials?			
	275	Are the ice packs in the freezer: • completely frozen? ☐ • partially frozen? ☐ • not frozen? ☐			
	276	Are the ice packs filled to the correct level (no leaks)?	Yes ☐	No ☐	
	277	Are they positioned correctly in the freezer?	Yes ☐	No ☐	

	278	Is there a functioning temperature monitoring device in the freezer?	Yes ☐	No ☐	If NO, go to question 280
	279	If YES, what is the temperature of the freezer?			
	280	Is there a monitoring chart on the freezer?	Yes ☐	No ☐	If NO, go to question 282
	281	If YES, since when?			
	282	Is there an IRL temperature monitoring device?	Yes ☐	No ☐	If NO, go to question 284
	283	If YES, what is the temperature inside?			
	284	Is there a monitoring chart on the IRL?	Yes ☐	No ☐	If NO, go to question 286
	285	If YES, how many times a day is the temperature recorded on the monitoring sheet?			
	286	Was there a failure in the temperature recording this year?	Yes ☐	No ☐	If NO, go to question 288
	287	If YES, when?			
	288	Have temperatures been outside the 2–8°C range this year (lower or higher)?	Yes ☐	No ☐	If NO, go to question 290
	289	If YES, when and what was the temperature?			
	290	Are there monitoring charts for the year of vaccination of confirmed cases (REFER TO QUESTION 8)?	Yes ☐	No ☐	If NO, go to question 293
	291	If YES, was there a failure in the temperature record that year?	Yes ☐	No ☐	If NO, go to question 293
	292	If YES, when?			
	293	Were temperatures outside the 2–8°C range that year (lower or higher)?	Yes ☐	No ☐	If NO, go to question 295
	294	If YES, when and what was the temperature?			
	295	Are the solvents kept at the right temperature according to the vaccine manufacturer?	Yes ☐	No ☐	
OBS HC	296	ASK THE PERSON IN CHARGE OF THE VACCINE STORE IF IT IS POSSIBLE TO LOOK AT THE PLACE WHERE THE VACCINES ARE STORED. FILL OUT THE QUESTIONNAIRES BELOW BASED ON THE OBSERVATIONS MADE. Are the yellow fever vaccines stored properly in the IRL?	Yes ☐	No ☐	
	297	Are there any yellow fever vaccines with the vaccine vial monitor at stage 3 or 4?	Yes ☐	No ☐	If NO, go to question 299
	298	If YES, how many vials?			
	299	Are there any expired vials?	Yes ☐	No ☐	If NO, go to question 301
	300	If YES, how many?			
	301	Are the ice packs in the freezer: • completely frozen? ☐ • partially frozen? ☐ • not frozen? ☐			
	302	Are the ice packs filled to the correct level (no leaks)?	Yes ☐	No ☐	
	303	Are they positioned correctly in the freezer?	Yes ☐	No ☐	
	304	Is there a functioning temperature monitoring device in the freezer?	Yes ☐	No ☐	If NO, go to question 306

	305	If YES, what is the temperature of the freezer?			
	306	Is there a monitoring chart on the freezer?	Yes ☐	No ☐	If NO, go to question 308
	307	If YES, since when?			
	308	Is there an IRL temperature monitoring device?	Yes ☐	No ☐	If NO, go to question 310
	309	If YES, what is the temperature inside the IRL?			
	310	Is there a monitoring chart on the IRL?	Yes ☐	No ☐	If NO, go to question 312
	311	If YES, how many times a day is the temperature recorded on the monitoring sheet?			
	312	Was there a failure in the temperature recording this year?	Yes ☐	No ☐	If NO, go to question 314
	313	If YES, when?			
	314	Have temperatures been outside the 2–8°C range this year (lower or higher)?	Yes ☐	No ☐	If NO, go to question 316
	315	If YES, when and what was the temperature?			
	316	Are there monitoring charts for the year of vaccination of confirmed cases (REFER TO QUESTION 8)?	Yes ☐	No ☐	If NO, go to question 319
	317	If YES, was there a failure in the temperature record that year?	Yes ☐	No ☐	If NO, go to question 319
	318	If YES, when?			
	319	Were temperatures outside the 2–8°C range that year (lower or higher)?	Yes ☐	No ☐	If NO, go to question 321
	320	If YES, when and what was the temperature?			
	321	Are the solvents kept at the right temperature according to the vaccine manufacturer?			
SUMMARY	***322***	***REVIEW THE RESPONSES TO QUESTIONS 224 TO 321.*** ***WAS THE VACCINE FAILURE CAUSED BY LOW-QUALITY YELLOW FEVER VACCINE ADMINISTRATION?***	Yes ☐	No ☐	***If NO, go to question 363***
	323	***IF YES, WAS THE LOW-QUALITY OF THE YELLOW FEVER VACCINES DUE TO:*** • ***A DEFECTIVE IRL?*** ☐ • ***POOR STORAGE OF THE YELLOW FEVER VACCINES AND/OR DILUENTS DURING AND OUTSIDE THE VACCINATION SESSIONS?*** ☐			
	324	***IF YES, AT WHICH LEVEL DID THE GAP OCCUR?*** • ***CENTRAL LEVEL?*** ☐ • ***DISTRICT LEVEL?*** ☐ • ***HEALTH CENTRE LEVEL?*** ☐			

Lack of trained staff

Data sources for finding information:

Interview with the logistics managers of the vaccine storage room at the central, district and health centre level (health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported)

Log C	325	Is there a dedicated person responsible for cold chain management at the central level?	Yes ☐	No ☐	If NO, go to question 329
	326	If YES, since when?			
	327	If YES, has this person been trained in cold chain and vaccine management?	Yes ☐	No ☐	If NO, go to question 329
	328	If YES, when was the last training?			
LOG D	329	Is there a dedicated person responsible for cold chain management at the district level?	Yes ☐	No ☐	If NO, go to question 333
	330	If YES, since when?			
	331	If YES, has this person been trained in cold chain and vaccine management?	Yes ☐	No ☐	If NO, go to question 333
	332	If YES, when was the last training?			
Log HC	333	Is there a dedicated person responsible for cold chain management at the health centre level?	Yes ☐	No ☐	If NO, go to question 337
	334	If YES, since when?			
	335	If YES, has this person been trained in cold chain and vaccine management?	Yes ☐	No ☐	If NO, go to question 337
	336	If YES, when was the last training?			

Misconceptions - lack of knowledge

Data sources for finding information:

Interview with the logistics managers of the vaccine storage room at the central, district and health centre level (health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported)

Observation of a routine vaccination session

EPI HC	337	Do you know at what temperature the yellow fever vaccine should be stored in the refrigerator?	Yes ☐	No ☐	If NO, go to question 339
	338	If YES, what temperature?			
	339	Do you know how long the yellow fever vaccine should be kept in the vaccine carrier or cool box?	Yes ☐	No ☐	If NO, go to question 341
	340	If YES, how long should it be kept?			
	341	Under what condition(s) should the yellow fever vaccine be stored in the vaccine carrier or cool box?			
	342	After how many hours should the reconstituted yellow fever vaccine be discarded?			

Obs vacci	343	Does the vaccinator know how to pack the icepacks and prepare the cool boxes?	Yes ☐	No ☐	
	344	Does the vaccinator know how to read and interpret the vaccine vial monitor?	Yes ☐	No ☐	
	345	Does the vaccinator know how to avoid freezing vaccines during transport?	Yes ☐	No ☐	
SUMMARY	***346***	***REVIEW THE RESPONSES TO QUESTIONS 337 to 345. IF AT LEAST ONE OF THE QUESTIONS WAS ANSWERED 'NO', CONTINUE WITH QUESTION 347; OTHERWISE, GO TO QUESTION 363.***			
DR and EPI C	347	Are there standard operating procedures for the cold chain and vaccine management in the country?	Yes ☐	No ☐	If NO, go to question 350
	348	If YES, ask for a copy and describe.			
	349	If YES, since when and how often?			
	350	Is there training on cold chain and vaccine management: • for supervisors? Yes ☐ No ☐ • for vaccinators? Yes ☐ No ☐ • other (DESCRIBE)? Yes ☐ No ☐			
	351	If YES, since when and how often: • for supervisors? Yes ☐ No ☐ • for vaccinators? Yes ☐ No ☐ • other (DESCRIBE)? Yes ☐ No ☐			
EPI D	352	Are there standard operating procedures for the cold chain and vaccine management at district level?	Yes ☐	No ☐	If NO, go to question 354
	353	If YES, ask for a copy and describe.			
	354	Have you received training related to cold chain and vaccine management?	Yes ☐	No ☐	If NO, go to question 356
	355	If YES, when?			Go to question 357
	356	If NO, is there an agent in your centre who is trained in cold chain management?			
EPI HC	357	REFER TO QUESTION 347 TO SEE IF STANDARD OPERATING PROCEDURES OR GUIDELINES ON COLD CHAIN AND VACCINE MANAGEMENT EXIST AT CENTRAL LEVEL. If YES, is there a copy in the health centre?	Yes ☐	No ☐	If NO, go to question 359
	358	If YES, ask to see it and describe briefly.			
	359	Have you received training related to cold chain and vaccine management?	Yes ☐	No ☐	If NO, go to question 361
	360	If YES, when?			Go to question 362
	361	If NO, is there an agent in your centre who is trained in cold chain management?	Yes ☐	No ☐	
SUMMARY	***362***	***REVIEW THE RESPONSES TO QUESTIONS 325 TO 361. WAS THE GAP IDENTIFIED IN QUESTION 322 DUE TO: • LACK OF TRAINED WORKERS? ☐ • LACK OF KNOWLEDGE? ☐***			

B.3.1.2 Failure of staff practice in injecting the yellow fever vaccine

Data sources for finding information:

Interview with the vaccinators at the health centre level (health centre which actively participates in vaccination activities and serves the health area where most confirmed yellow fever cases were reported)

Observation of vaccination sessions

Standard operating procedures on yellow fever vaccine handling and reconstitution

EPI HC	363	Do you know how to reconstitute the yellow fever vaccine?	Yes ☐	No ☐	If NO, go to question 365
	364	If YES, please explain the reconstitution process.			
	365	What route do you use to administer the yellow fever vaccine? (IF POSSIBLE, OBSERVE THE ADMINISTRATION OF THE YELLOW FEVER VACCINE ON A ROUTINE VACCINATION DAY.) IF THE ANSWER TO THE PREVIOUS QUESTIONS ARE INCORRECT, PROCEED WITH THE FOLLOWING QUESTIONS, OTHERWISE GO TO Q 368.			
	366	Have you received training on vaccine handling practices?	Yes ☐	No ☐	If NO, go to question 368
	367	If YES, when?			
Obs vacci	368	Does the vaccinator use a diluent and a vaccine from the same manufacturer?	Yes ☐	No ☐	
	369	Are the diluents stored outside the vaccine carrier/cool box during transport?	Yes ☐	No ☐	
	370	Does the vaccinator inject the vaccine by subcutaneous cutaneous injection?	Yes ☐	No ☐	
SUMMARY	***371***	***REVIEW THE RESPONSES TO QUESTIONS 363 TO 370.*** ***WAS THE VACCINE FAILURE DUE TO:***			
		• ***A FAILURE IN INJECTING CORRECTLY THE YELLOW FEVER VACCINE?***	YES ☐	NO ☐	
		• ***A GAP IN RECONSTITUTING THE VACCINE?***	YES ☐	NO ☐	

B.3.2 User-based reasons

Data sources for finding information:

Published articles/studies regarding vaccine failure with yellow fever vaccine in the country

DR	372	IDENTIFY IF THERE ARE ANY STUDIES IN THE COUNTRY OR IN THE DISTRICT SHOWING PRIMARY OR SECONDARY YELLOW FEVER VACCINATION FAILURE.	Yes ☐	No ☐	If NO, go to question 374
	373	If YES, describe.			
SUMMARY	***374***	***REVIEW THE RESPONSES TO QUESTIONS 372 TO 373.*** ***WAS THERE A VACCINE FAILURE THAT COULD HAVE CONTRIBUTED TO THE YELLOW FEVER OUTBREAK?***	Yes ☐	No ☐	

References

1. Risk communication and community engagement readiness and response toolkit: yellow fever. Geneva: World Health Organization; 2024. (https://iris.who.int/handle/10665/376259). Licence: CC BY-NC-SA 3.0 IGO.
2. Weekly Epidemiological Record. 12 December 2014; 89 (50):561–576). Geneva. World Health Organization. (https://iris.who.int/items/32075c61-eb8f-49cc-8a11-592597011744).
3. Measles outbreak guide. Geneva: World Health Organization; 2022. (https://iris.who.int/handle/10665/360891). Licence: CC BY-NC-SA 3.0 IGO
4. Yellow fever: vaccine-preventable diseases surveillance standards. Geneva: World Health Organization; 2020. (https://www.who.int/publications/m/item/vaccine-preventable-diseases-surveillance-standards-yellow-fever).
5. Eliminate yellow fever epidemics (EYE) strategy country toolkit. Geneva: World Health Organization; 2024. (https://iris.who.int/handle/10665/379980). Licence: CC BY-NC-SA 3.0 IGO.

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